BLE Mesh based IR AC Controller

Software Design Document

Apr 10, 2024

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Revision History

Version	Release Date	Prepared by	Remarks
0.2	01.04.2024	Kulasekaran	• Draft Version
0.3	10.04.2024	Kulasekaran	• Removed the HeartBeat related content as temperature data has been decided to be taken as heartbeat • Added Teaching mode packet and ack for Gwy and Node

Acronyms and Abbreviations

MQTT	Message Queuing Telemetry Transport
BLE	Bluetooth Low Energy
UART	Universal Asynchronous Receiver / Transmitter
LTE	Long-term evolution (4G)
GWY	Gateway
IR	Infrared
ACK	Acknowledgement

Introduction

Unimation Robotics intends to develop a Bluetooth Mesh based system to control Air conditioners remotely. QMAX Systems India Pvt. Ltd. an Electronics Engineering and R&D Services Company will design, develop, test, and deliver prototypes matching the requirements for Unimation Robotics.

Purpose of Document

The purpose of this document is to describe the Software design details of the product being developed by QMAX Systems India Pvt. Ltd. for Unimation Robotics.

Product Brief

The overall product mainly focuses on building a BLE Mesh based system consisting of two types of devices: 1. Gateway 2. Node. Both the devices are capable of controlling Air conditioners by transmitting IR signals using the IR transmitter on-board. Also both the devices are capable of detecting IR signals using the IR receiver on-board. Adding to that, both the devices have temperature sensors on-board, that help in tracking the ambient temperature and do logical operation based on requirement. Gateway is master type device and Node is a slave type device. Gateways receive commands from cloud and perform necessary operation or relay them to the appropriate node to perform the necessary task. Nodes after performing the task will relay back acknowledgement to Gateway, which will be relayed back to Cloud.

Software Architecture

The following image represents the Software architecture of the whole system and how the devices communicate with each other.

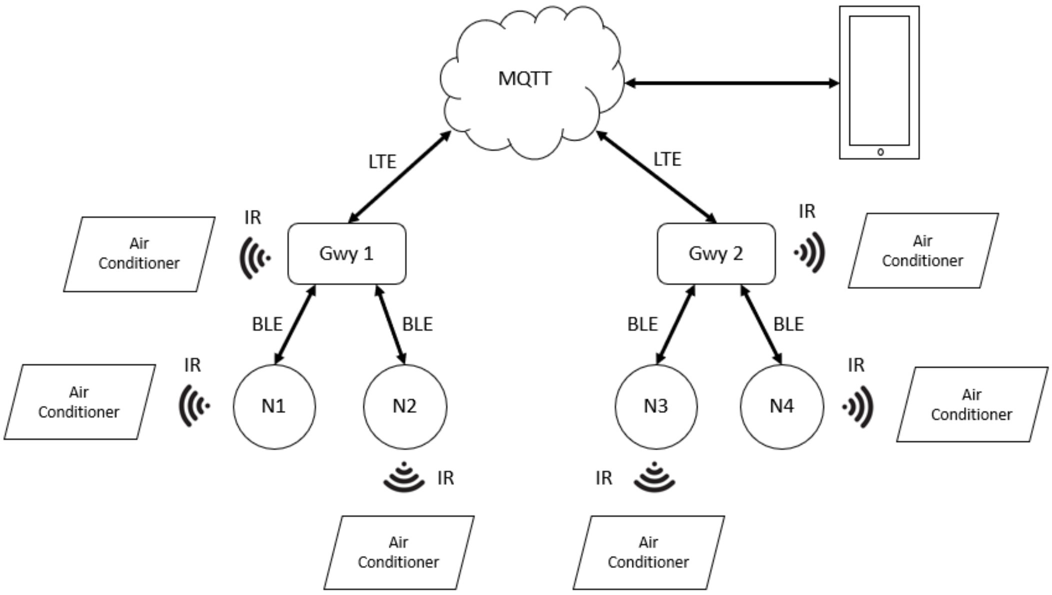


Figure 1 Software Architecture Block Diagram

Basic processes and Terminologies

BLE Mesh based systems involve a few initial steps to be followed inorder to add/remove devices into/from the network. The following section provides information reg. what those steps/processes are.

Central Server

Central server is the place from where the entire system receives commands and sends ACKs.

Gateway

Gateway is a type of device in the BLE Mesh network which act as a mediator between the slave devices (Nodes) and the central server. Each Gateway is identified the BLE Mesh network by their unique serial number.

Registration

Registration is the process of adding a Gateway device into the BLE Mesh network.

Unregistration

Unregistration is the process of removing a Gateway deivce from the BLE Mesh network.

Node

Node is a type of device in the BLE Mesh network which act as slaves. Typically, Nodes on a BLE Mesh network don’t communicate with the Central Server directly. They will communicate via Gateway to the Central Server.

Provisioning

Provisioning is the process of adding a Node device into the BLE Mesh network.

Unprovisioning

Unprovisioning is the process of removing a Node device from the BLE Mesh network.

MQTT Publishing

In MQTT, publishing is the concept of sending data to a specific topic. Topics are like channels in MQTT. There can be any number of topics and all topics are unique. Devices that have subscribed to the topic will receive data from that topic.

MQTT Subscribing

In MQTT, Subscribing is the concept of listening for data from a specific topic.

Gateway LED Indications

Table 0.1 Gateway LED Indication

Sl.No	Pattern	Description
1	Toggle Red + Green + Blue	Gateway is in MQTT configuration mode.
2	Solid Red	Gateway is not connected with MQTT server.
3	Blinking Red + Blue	Gateway is not registered.
4	Slow Blinking Blue (500ms ON | 500ms OFF)	Gateway is in AC remote configuration mode.
5	Fast Blinking Blue (200ms ON | 200ms OFF)	Gateway is in Teaching mode.
6	Solid Green	Gateway is connected with MQTT, Registered and Configured. Meaning, the Gateway is fully operational.

Node LED Indications

Table 0.2 Node LED Indication

Sl.No	Parameter	Description
1	Blinking Red + Blue	Node is not Provisioned.
2	Slow Blinking Blue (500ms ON | 500ms OFF)	Node is in AC remote configuration mode.
3	Fast Blinking Blue (200ms ON | 200ms OFF)	Node is in Teaching mode.
4	Solid Green	Node is provisioned and configured. Meaning, the node is fully operational.

MQTT Communication Packets

The communication between Gateway and MQTT cloud is based on JSON packets. The communication can take place in both directions i.e., from the gateway to the cloud and vice-versa.

Note: If a packet name has “Ack” at its end, then it means, the origin of the packet is from the device’s side. All other packets, have origin from the Cloud end.

Table 0.3 MQTT Communication Packets

Packet Name	Packet ID	Packet Description
GATEWAY MQTT PACKETS		
Gateway Registration	0	Adds a Gwy into BLE Mesh network
Gateway AC Remote Configuration Ack	1	Sent back as ACK to cloud. Contains information about Gateway AC remote configuration
Gateway Unregistration	2	Removes a Gwy from BLE Mesh network
Gateway AC Control	3	Contains AC control settings to control AC using Gwy
Gateway AC manual control Ack	4	Sent back as ACK to cloud. Contains information about manual control of AC using remote
Gateway AC Remote Reconfiguration	5	Puts Gwy into AC remote configuration mode
Gateway Temperature data Ack	6	Sent back as ACK to cloud. Contains ambient temperature value measured by the on-board Temperature sensor on Gwy. This data acts as a Heartbeat as well.
Gateway Temperature Publish configuration	7	Tells the Gwy, how frequently it needs to send the Temperature data ACK to cloud
Gateway Reset MQTT	8	Disconnects Gateway from MQTT network. Resets MQTT parameters in Gwy and puts the Gwy into MQTT configuration mode.
Gateway Teaching Mode Start	9	Puts the Gateway in AC remote teaching mode
Gateway Teaching Mode End ACK	10	Sent back as ACK to cloud. Tells if the Gwy has exited Teaching mode or not.
NODE MQTT PACKETS		
Node Provisioning	100	Adds a Node into BLE Mesh network
Node AC Remote Configuration Ack	101	Sent back as ACK to cloud. Contains information about Node AC remote configuration
Node Unprovisioning	102	Removes a Node from BLE Mesh network
Node AC control	103	Contains AC control settings to control AC using Node
Node AC manual control Ack	104	Sent back as ACK to cloud. Contains information about manual control of AC using remote

Node AC Remote Reconfiguration	105	Puts Node into Configuration mode
Node Temperature data Ack	106	Sent back as ACK to cloud. Contains ambient temperature value measured by the on-board Temperature sensor on Node
Node Temperature Publish configuration	107	Tells the Node, how frequently it needs to send the Temperature data ACK to Gwy
Node Teaching Mode	108	Puts a Node into AC Remote teaching mode.
Node Teachin Mode End ACK	109	Sent back as ACK to cloud. Tells if the Node has exited Teaching mode or not.

Error codes

Below table contains all the error codes used across the application and what they mean.

Table 0.4 Error Codes

Error code	Error code name	Description / Use Case
	GENERAL	
-1	FAILURE	Operation has failed
0	SUCCESS	Operation has succeeded
1	INVALID_JSON_PACKET_ID	Unknown JSON Packet ID
2	INAVLID_MSG_SEQ_NO	Invalid Msg Sequence No
3	INVALID_GWY_SER_NO	Invalid Gwy Serial No
4	INVALID_NODE_SER_NO	Invalid Node Serial No
5	INVALID_LOCATION_STR	Invalid Location
6	NODE_COMM_TIMEOUT	Gwy has failed to connect with Node
	GWY REG & UNREG	
100	GWY_ALREADY_REG	User is trying to register a Gwy that is already registered
101	GWY_ALREADY_UNREG	User is trying to Unregister a Gwy that is not yet registered
	NODE PROV & UNPROV	
200	NODE_ALREADY_PROV	User is trying to provision a node that is already provisioned

201	NODE_ALREADY_UNPROV	User is trying to Unprovision a node that is already provisioned
CONFIG & RECONFIG		
300	GWY_ALREADY_AC_REMOTE_UNCONF	User is trying to unconfigure a Gwy that has not been configured
301	NODE_ALREADY_AC_REMOTE_UNCONF	User trying to unconfigure a Node that has not been configured
AC CONTROL		
400	GWY_NOT_REG	User is trying to control AC with a Gwy that has not been registered
401	GWY_NOT_AC_REMOTE_CONF	User is trying to control AC with a Gwy that has not been AC remote configured
402	INVALID_POWER	Invalid Power
403	INVALID_MODE	Invalid Mode
404	INVALID_FAN_SPEED	Invalid Fan speed
405	INVALID_TEMPERATURE	Invalid Temperature
406	INVALID_SWING_H	Invalid Horizontal Swing
407	INVALID_SWING_V	Invalid Vertical Swing
408	INVALID_ONTIMER	Invalid On timer
409	INVALID_OFFTIMER	Invalid Off timer
410	INVALID_LOCKING	Invalid Locking
411	INVALID_TEMP_UPPER_LIMIT	Invalid Upper Temperature limit
412	INVALID_TEMP_LOWER_LIMIT	Invalid Lower Temperature limit
413	LOCKING_TEMP_UP_LIMIT_EXCEEDING_TEMP_UP_LIMIT	User set locking temperature up limit is more than the the absolute temperature up limit
414	LOCKING_TEMP_LOW_LIMIT_EXCEEDIGN_TEMP_LOW_LIMIT	User set locking temperature low limit is lower than the absolute temperature low limit

415	ILLOGICAL_LOCKING_TEMP_LIMIT	TempLockUpLimit is lower than TempLockLowLimit
416	EXCEEDING_TEMP_LOWER_LIMIT	User set temperature value is exceeding the absolute temperature low limit.
417	EXCEEDING_TEMP_UPPER_LIMIT	User set temperature value is exceeding the absolute temperature up limit
	HEARTBEAT	
500	INVALID_PUBLISH_PERIOD	
	MISCELLANEOUS	
999	UNKNOWN_ERROR	

Common JSON parameters

The following parameters are used across most of the MQTT communication packets. Below table provides information about these parameters and the purpose behind it.

Table 0.5 Common JSON Parameter description

Sl.No	Parameter	Description
1	JsonPacketId	This is used for identification on both cloud side and Gwy side to know which packet is being received
2	MsgSeqNo	The message sequence number is used to identify which ACK corresponds to which command.
3	GwySerNo	Which Gateway device, the command is intended for
4	Error Code	Included on all ACK packets. Lets the cloud know if the operation has succeeded or needs attention.
5	Location	This is used to identify the physical location of where the device was installed. User sets this information upon Registration / Provision process.

Gateway Registration Packet

Description : This packet is used to add a Gwy device into the BLE Mesh network.

LED Indication: If Unregistered, Gateway blinks RED + BLUE.

Gateway Registration JSON (Cloud to Gateway):

```
{                                                                    Range
    "JsonPacketId"  :  0            // (Number)
    "MsgSeqNo"      :  12           // (Number)                  [0 - 65535]
    "GwySerNo"      :  "GWY00001"   // (String)                [00001 - 99999]
    "Location"      :  "1st Floor"  // (String)                [20 chars long]
}
```

Process flow diagram :

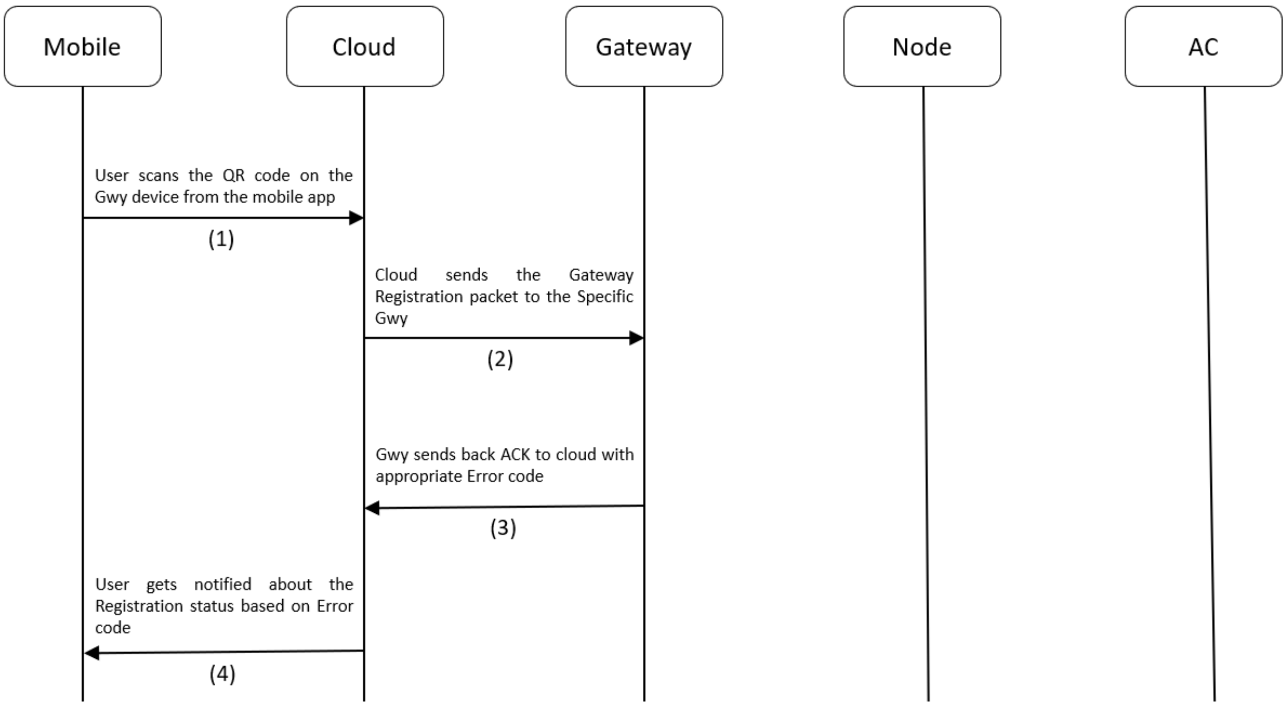


Figure 2 Gateway Registration process flow

Gateway Registration Ack

Description : This ack is used by cloud to know whether the Gwy registration was successful or not. Gateway sends back this packet after receiving Gateway Registration packet.

Gateway Registration JSON ACK (Gateway to Cloud):

```
{                                                                              Range
    "JsonPacketId"   :  0              // (Number)
    "MsgSeqNo"       :  12             // (Number)                         [0 - 65535]
    "GwySerNo"       :  "GWY00001"     // (String)                       [00001 - 99999]
    "Location"       :  "1st Floor"    // (String)                       [20 chars long]
    "ErrorCode"      :  0              // (Number)
}
```

Process flow diagram :

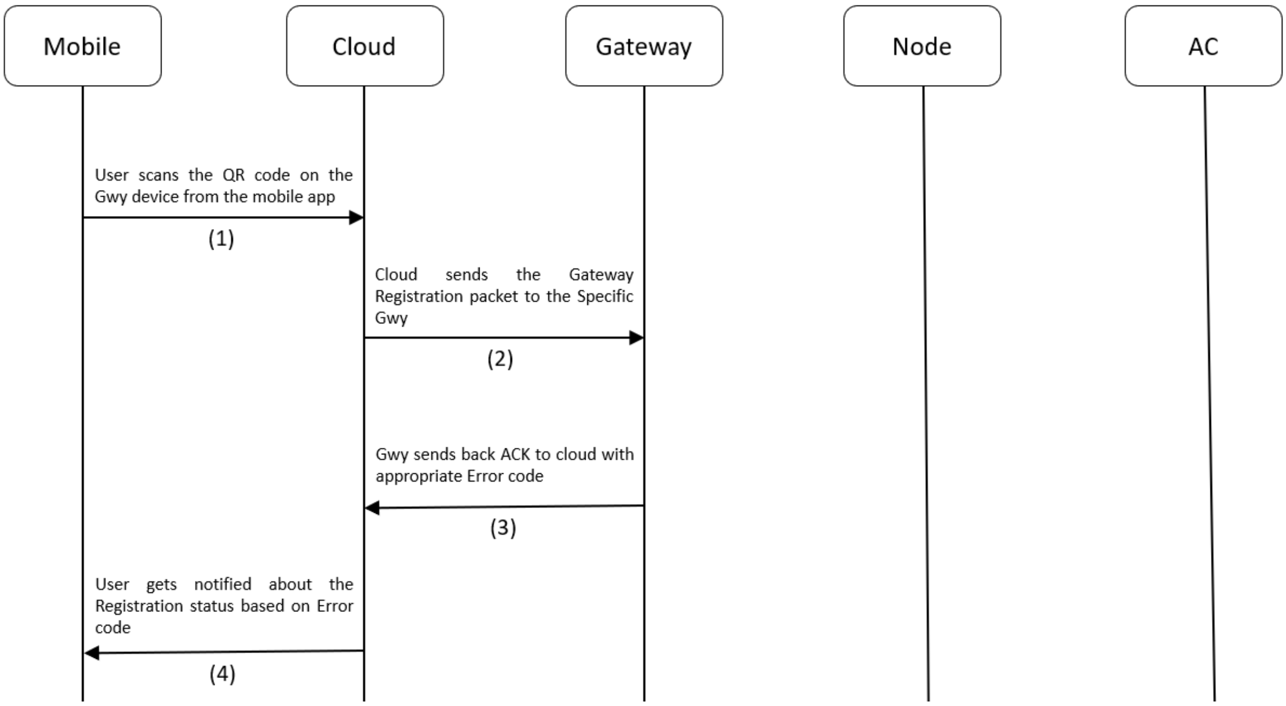


Figure 3 Gateway Registration process flow

Gateway AC Remote Configuration Ack

Description : This is an ACK sent by Gwy to Cloud when the Gwy is successfully configured with an AC remote. When Gwy is in AC Remote Configuration mode (Slow Blinking BLUE light), and AC remote is pressed in-front of the IR receiver in the Gwy, it will get configured to act as that AC remote from there on and send this ACK to the cloud.

LED Indication: If not configured, Gateway will be slow blinking BLUE (500ms ON | 500ms OFF), meaning the Gateway is in AC Remote configuration mode.

Gateway AC Remote Configuration JSON ACK (Gateway to Cloud):

```
{                                                                     Range
    “JsonPacketId”  :  1               // (Number)
    “GwySerNo”      :  “GWY00001”      // (String)             [00001 - 99999]
    “Location”      :  “1st Floor”     // (String)             [20 chars long]
    “ErrorCode”     :  0               // (Number)
}
```

Process flow :

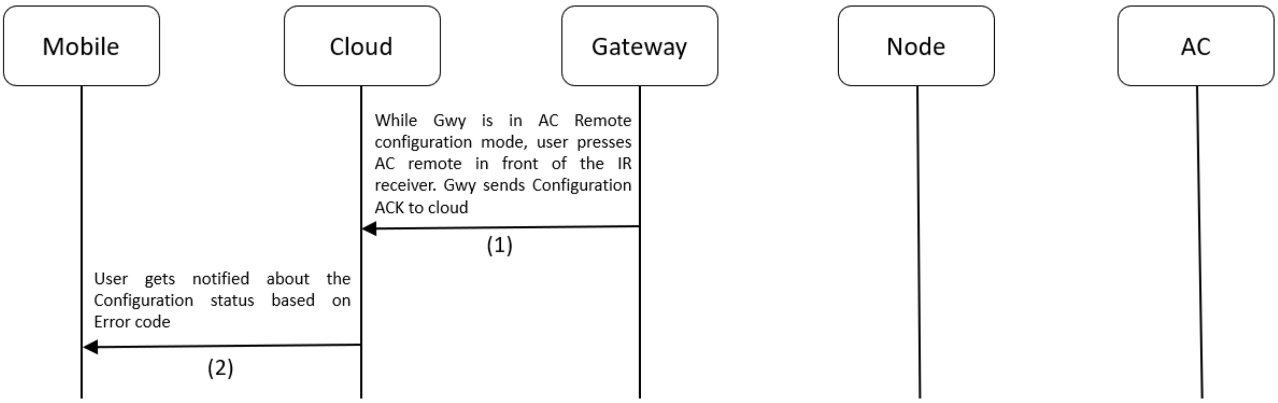


Figure 4 Gateway Configuration process flow

Gateway Unregistration Packet

Description : This packet is used to remove a Gwy device from the BLE Mesh network.

Note: When a Gateway is unregistered, it will also become unconfigured automatically.

Gateway Unregistration JSON (Cloud to Gateway):

				Range
{				
"JsonPacketId"	:	2	// (Number)	
"MsgSeqNo"	:	12	// (Number)	[0 - 65535]
"GwySerNo"	:	"GWY00001"	// (String)	[00001 - 99999]
"Location"	:	"1st Floor"	// (String)	[20 chars long]
}				

Process flow :

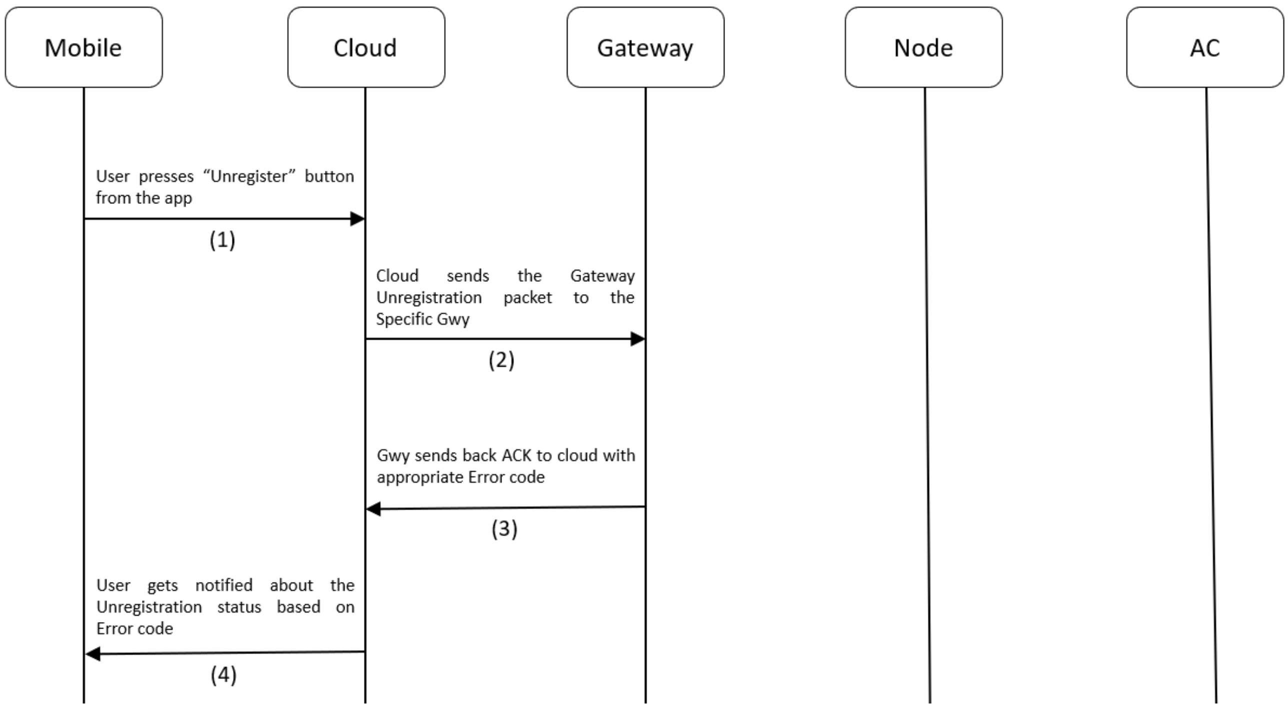


Figure 5 Gateway Unregistration Process flow

Gateway Unregistration Ack

Description : This ACK is used by cloud to know whether the Gwy Unregistration was successful or not. This packet is sent by Gateway after receiving Gateway Unregistration packet.

Gateway Unregisration JSON ACK (Gateway to Cloud):

```
{
    "JsonPacketId"  :  2            // (Number)
    "MsgSeqNo"      :  12           // (Number)
    "GwySerNo"      :  "GWY00001"   // (String)
    "Location"      :  "1st Floor"  // (String)
    "ErrorCode"     :  0            // (Number)
}
```

Process flow :

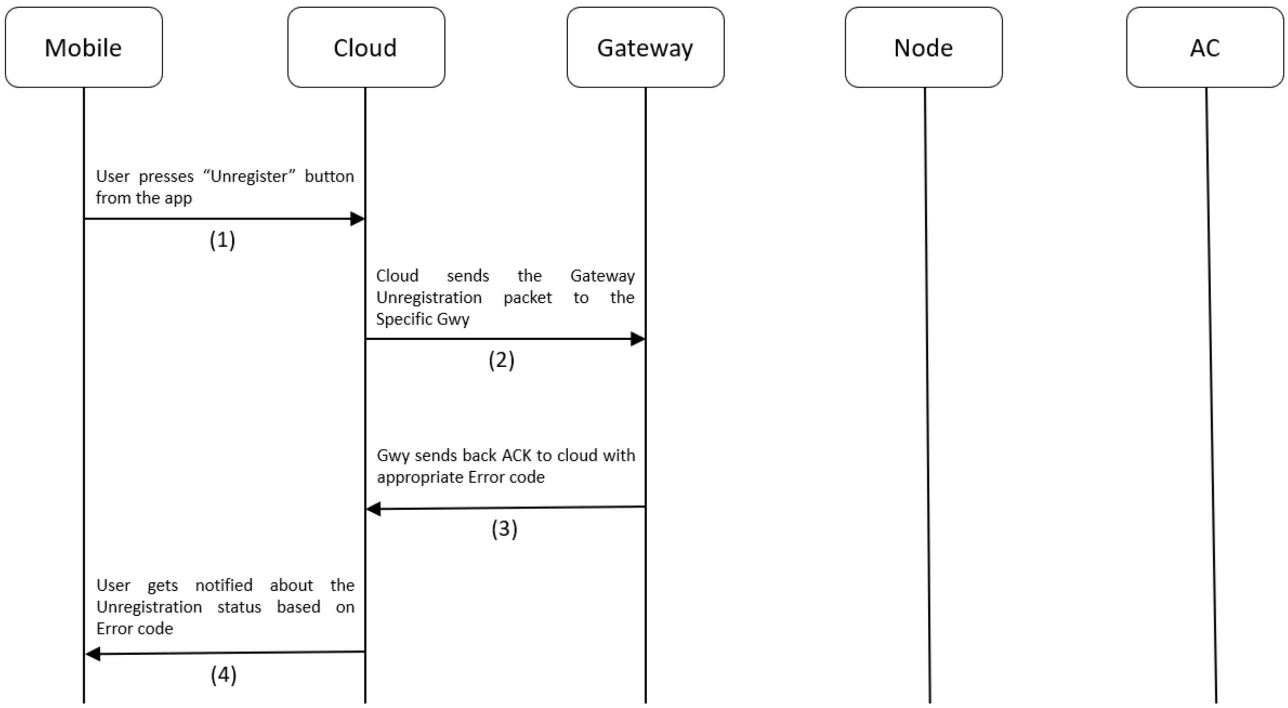


Figure 6 Gateway Unregistration Process Flow

Gateway AC Control Packet

Description: This packet is used to control an AC using the Gwy.

Sl.No	Parameter	Description	Possible Err Codes
1	Power	1 = ON, 0 = OFF	INVALID_POWER
2	Mode	Following modes are supported : [Cool, Dry, Hot, Fan, Auto].	INVALID_MODE
3	FanSpeed	0 = AUTO, 1 – 5 Levels	INVALID_FAN_SPEED
3	Temperature	Sets the AC temperature. Must be within 18 and 32.	EXCEEDING_TEMP_LOWER_LIMIT, EXCEEDING_TEMP_UPPER_LIMIT
4	SwingH	1 = ON, 0 = OFF	INVALID_SWING_H
5	SwingV	1 = ON, 0 = OFF	INVALID_SWING_V
6	OnTimer	Sets the number of hours past midnight after which AC needs to be turned on automatically	INVALID_ONTIMER
7	OffTimer	Sets the number of hours past midnight after which AC needs to be turned off automatically	INVALID_OFFTIMER
8	Locking	1 = ON, 0 = OFF. When ON, if AC was controlled manually, then Locking ACK will be sent to Cloud. Also, when ON, the AC will be maintained between the TempLockUpLimit and TempLockLowLimit. If by manual control, AC temp was set exceeding the above limits, then AC's temperature will be brought back under accepted limits.	INVALID_LOCKING
9	TempLockUpLimit	Specifies upper temperature locking limit. This parameter will have no effect application wise, when Locking is not enabled. Must have values within the absolute Temperature limits : 18 - 32. Default value : 27	INVALID_TEMP_UPPER_LIMIT, ILLOGICAL_LOCKING_TEMP_LIMIT

10	TempLockLowLimit	Specifies Lower temperature locking limit. This parameter will have no effect application wise, when Locking is not enabled. Must have values within the absolute Temperature limits : 18 - 32. Default value : 20.	INVALID_TEMP_LOWER_LIMIT, ILLOGICAL_LOCKING_TEMP_LIMIT

Gateway AC Control JSON packet (Cloud to Gateway):

```
{                                                                                 Range
    "JsonPacketId"        :   3              // (Number)
    "MsgSeqNo"            :   12             // (Number)     [0 - 65535]
    "GwySerNo"            :   "GWY00001"     // (String)     [00001 - 99999]
    "Power"               :   1              // (Number)     [0 = OFF, 1 = ON]
    "Mode"                :   "Cool"         // (String)     ["Cool", "Dry", "Auto", "Fan", "Hot"]
    "FanSpeed"            :   1              // (Number)     [0 - 5]
    "Temperature"         :   25             // (Number)     [18 - 32]
    "SwingH"              :   1              // (Number)     [0 = OFF, 1 = ON]
    "SwingV"              :   1              // (Number)     [0 = OFF, 1 = ON]
    "OnTimer"             :   0              // (Number)     [0 - 12]
    "OffTimer"            :   0              // (Number)     [0 - 12]
    "Locking"             :   1              // (Number)     [0 = OFF, 1 = ON]
    "TempLockUpLimit"     :   26             // (Number)     [18 - 32]
    "TempLockLowLimit"    :   20             // (Number)     [18 - 32]
}
```

Process flow :

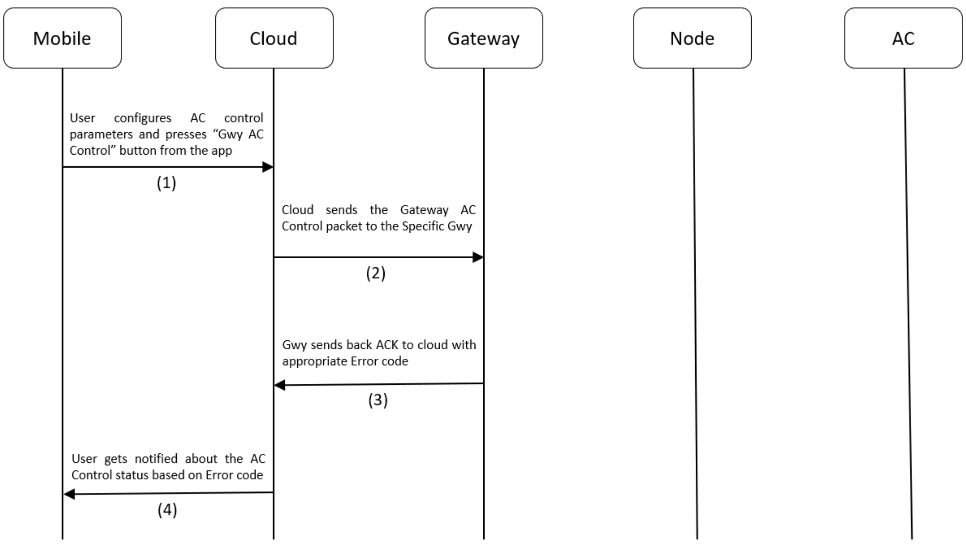


Figure 7 Gateway AC Control Process flow

Gateway AC Control Ack

Description: This ack is used by cloud to know whether the Gwy AC Control was successful or not.

Gateway AC Control JSON ACK (Gateway to Cloud):


```
{
    "JsonPacketId"       :  3            // (Number)
    "MsgSeqNo"           :  12           // (Number)
    "GwySerNo"           :  "GWY00001"   // (String)
    "Power"              :  1            // (Number)
    "Mode"               :  "Cool"       // (String)
    "FanSpeed"           :  1            // (Number)
    "Temperature"        :  25           // (Number)
    "SwingH"             :  1            // (Number)
    "SwingV"             :  1            // (Number)
    "OnTimer"            :  0            // (Number)
    "OffTimer"           :  0            // (Number)
    "Locking"            :  1            // (Number)
    "TempLockUpLimit"    :  26           // (Number)
    "TempLockLowLimit"   :  20           // (Number)
    "ErrorCode"          :  0            // (Number)
}
```


Process flow :

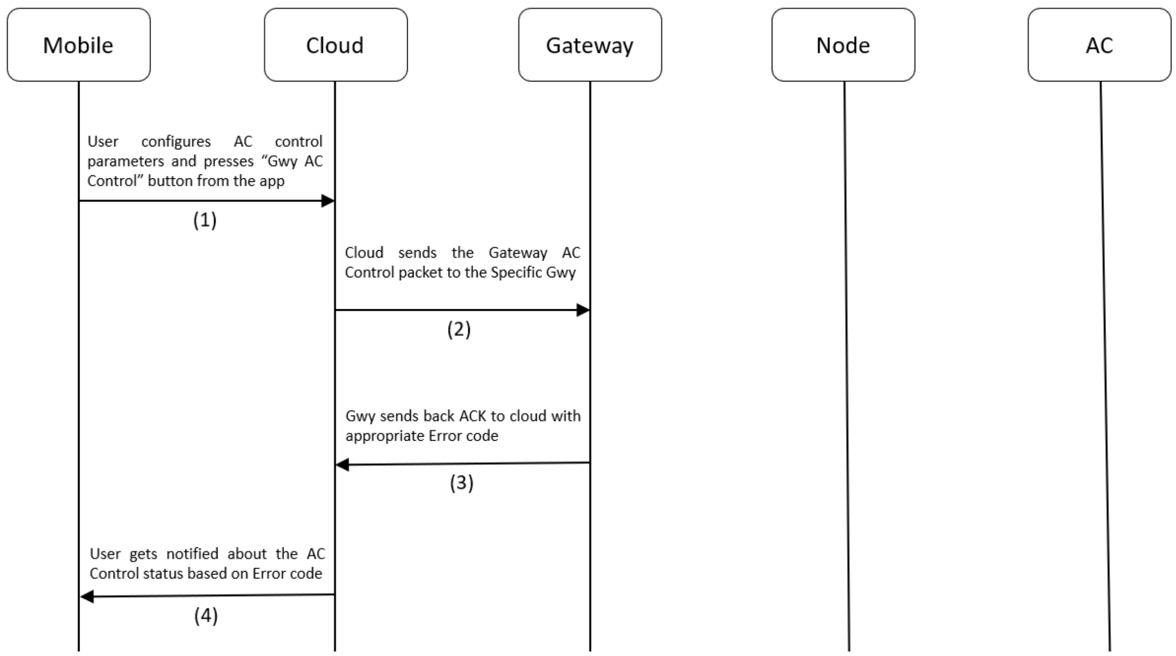


Figure 8 Gateway AC Control Process flow

Gateway AC manual control Ack

Description: This ack is used by cloud to know whether the AC was controlled manually using the AC remote. If someone controls the AC manually using the remote, then the IR receiver on-board, will capture this IR signal and decode it. If it's a signal from the same remote with which the Gateway was configured with during the AC Remote configuration process, then it will store the manual control information for sending it to the cloud. Also it performs logical operation based on the temperature that was set using the remote. If it exceeds the Locking temperature limits set from the AC Control packet, then it will bring the AC's temperature back to permissible limits.
Note: This ACK will be sent from Gateway only if the Locking feature had been enabled. By default, the TempLockUpLimit is set to 27 and TempLockLowLimit is set to 20.

Gateway AC Locking JSON ACK (Gateway to Cloud):

```
{
    "JsonPacketId"      :  4              // (Number)
    "MsgSeqNo"          :  12             // (Number)
    "GwySerNo"          :  "GWY00001"     // (String)
    "Power"             :  1              // (Number)
    "Mode"              :  "Cool"         // (String)
    "FanSpeed"          :  1              // (Number)
    "Temperature"       :  25             // (Number)
    "SwingH"            :  1              // (Number)
    "SwingV"            :  1              // (Number)
    "OnTimer"           :  0              // (Number)
    "OffTimer"          :  0              // (Number)
}
```

Process flow :

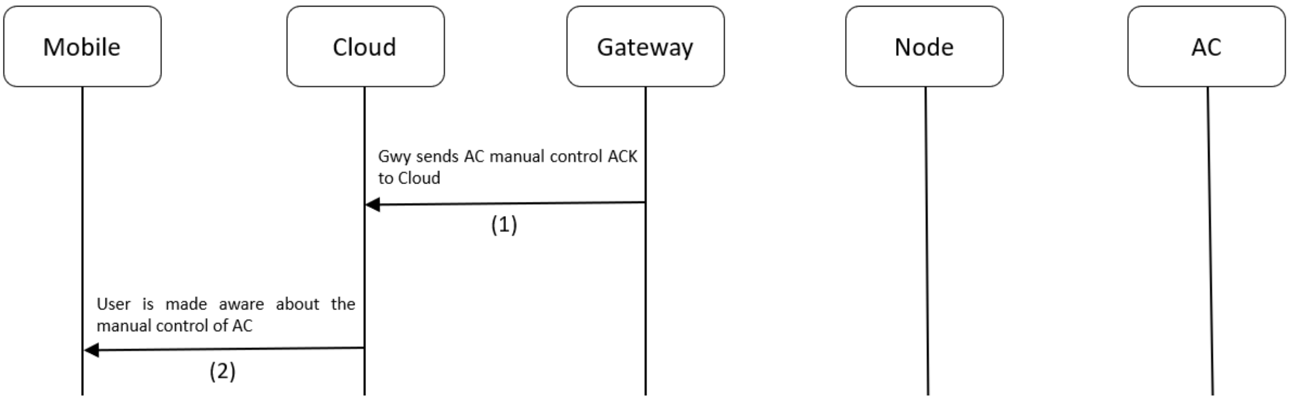


Figure 9 Gateway AC manual control Process flow

Gateway AC Remote Reconfiguration Packet

Description: This packet is used to put the Gwy in AC remote configuration mode.

Gateway AC Remote Reconfiguration JSON (Cloud to Gateway):

```
{                                                                              Range
    “JsonPacketId”   :   5              // (Number)
    “MsgSeqNo”       :   12             // (Number)                      [0 - 65535]
    “GwySerNo”       :   “GWY00001”     // (String)                   [00001 - 99999]
}
```

Process flow :

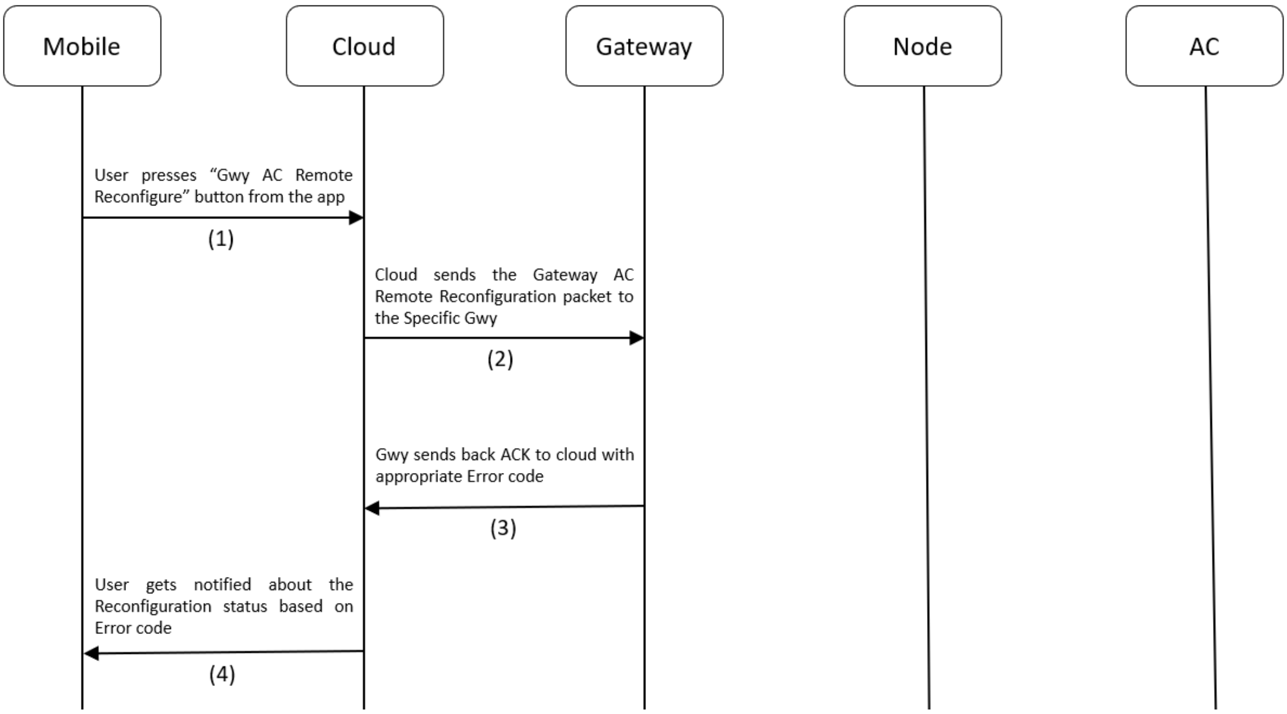


Figure 10 Gateway AC Remote Reconfiguration Process Flow

Gateway AC Remote Reconfiguration Ack

Description: This ACK is used by cloud to know if the Gwy has entered AC Remote reconfiguration mode after receiving the reconfiguration packet.

Gateway AC Remote Reconfiguration JSON ACK (Gateway to Cloud):

```
{
    "JsonPacketId"   :  5            // (Number)
    "MsgSeqNo"       :  12           // (Number)
    "GwySerNo"       :  "GWY00001"   // (String)
    "ErrorCode"      :  0            // (Number)
}
```

Process flow :

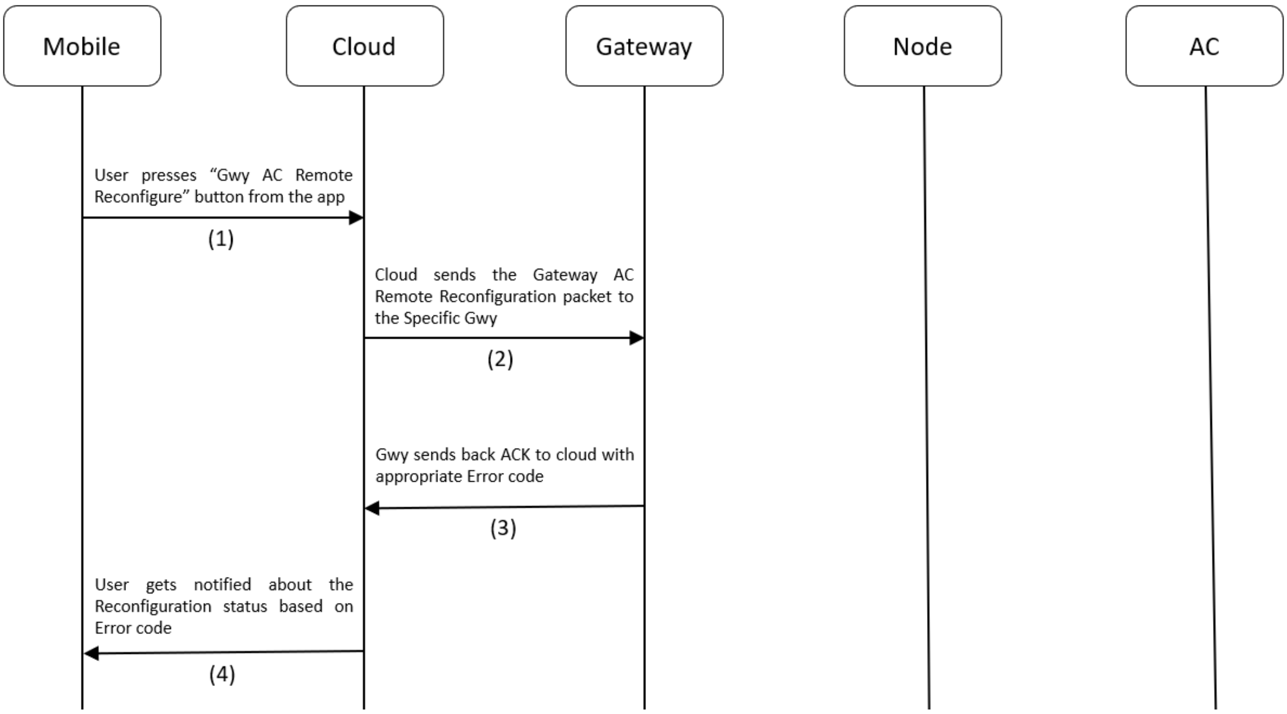


Figure 11 Gateway AC Remote Reconfiguration Process flow

Gateway Temperature Data Ack

Description: This ACK is used by cloud to know the ambient temperature of the environment in which the Gwy is operating. This ACK is sent periodically to the cloud. This frequency can be configured in the Gateway Temperature Publish Configuration Packet.

Note: This ACK is used for Heartbeat purpose as well.

Gateway Temperature Data JSON ACK (Gateway to Cloud):

```
{
    "JsonPacketId"   :  6             // (Number)
    "MsgSeqNo"       :  12            // (Number)
    "GwySerNo"       :  "GWY00001"    // (String)
    "Temperature"    :  23            // (Number)
}
```

Process flow :

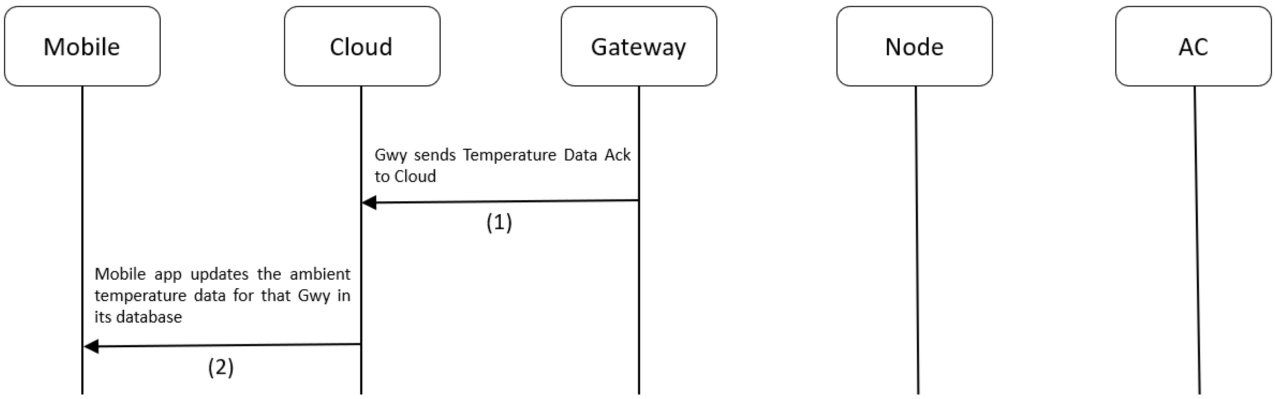


Figure 12 Gateway Temperature Data Process flow

Gateway Temperature Publish configuration Packet

Description: This packet is used to tell the Gwy, how frequently it needs to send the Temperature data ACK to cloud. The frequency is configurable in seconds.

Gateway Temperature Publish configuration JSON (Cloud to Gateway):


```
{                                                                              Range
    "JsonPacketId"      :  7                 // (Number)
    "MsgSeqNo"          :  12                // (Number)                   [0 – 65535]
    "GwySerNo"          :  "GWY00001"        // (String)                   [00000 - 99999]
    "PublishPeriodSec"  :  20                // (Number) (in Sec)          [0 - 65535]
}
```


Process flow :

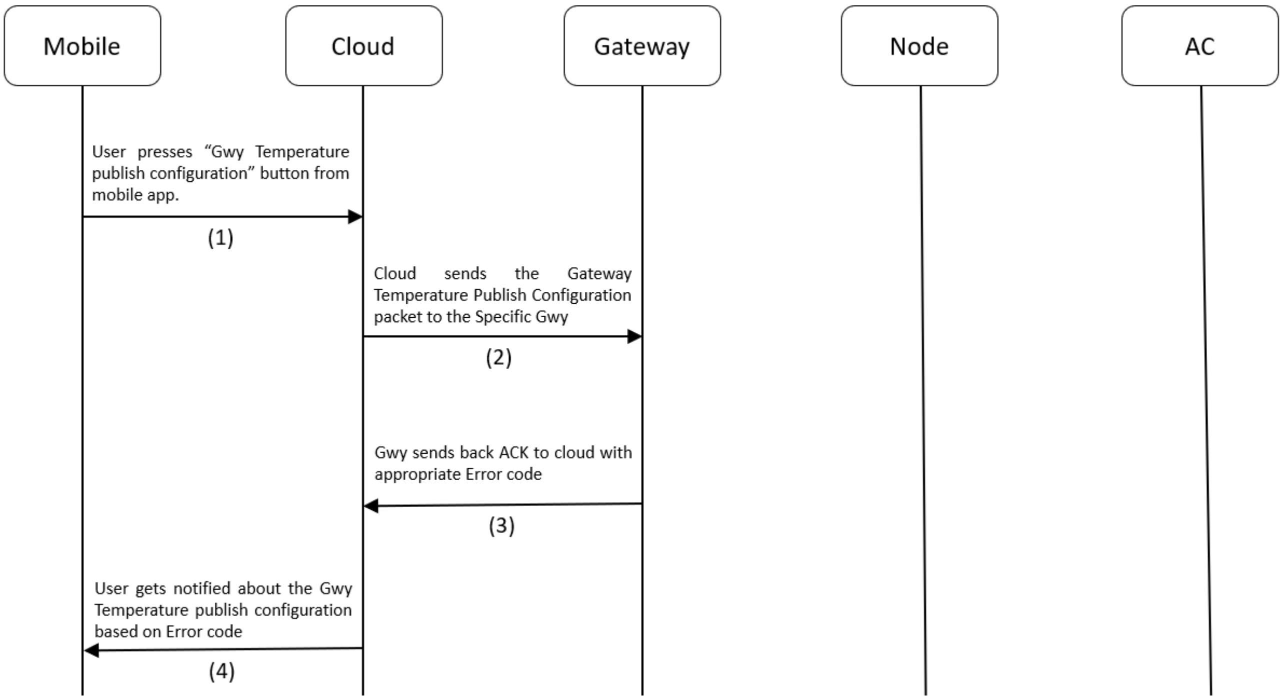


Figure 13 Gateway Publish Configuration Process flow

Gateway Temperature Publish configuration Ack

Description: This ACK is used by cloud to know whether the Publish configuration was successful or not.

Gateway Temperature Publish Configuration JSON ACK (Gateway to Cloud):


```
{
    "JsonPacketId"      :  7              // (Number)
    "MsgSeqNo"          :  12             // (Number)
    "GwySerNo"          :  "GWY00001"     // (String)
    "PublishPeriodSec"  :  20             // (Number) (in Sec)
    "ErrorCode"         :  0              // (Number)
}
```


Process flow :

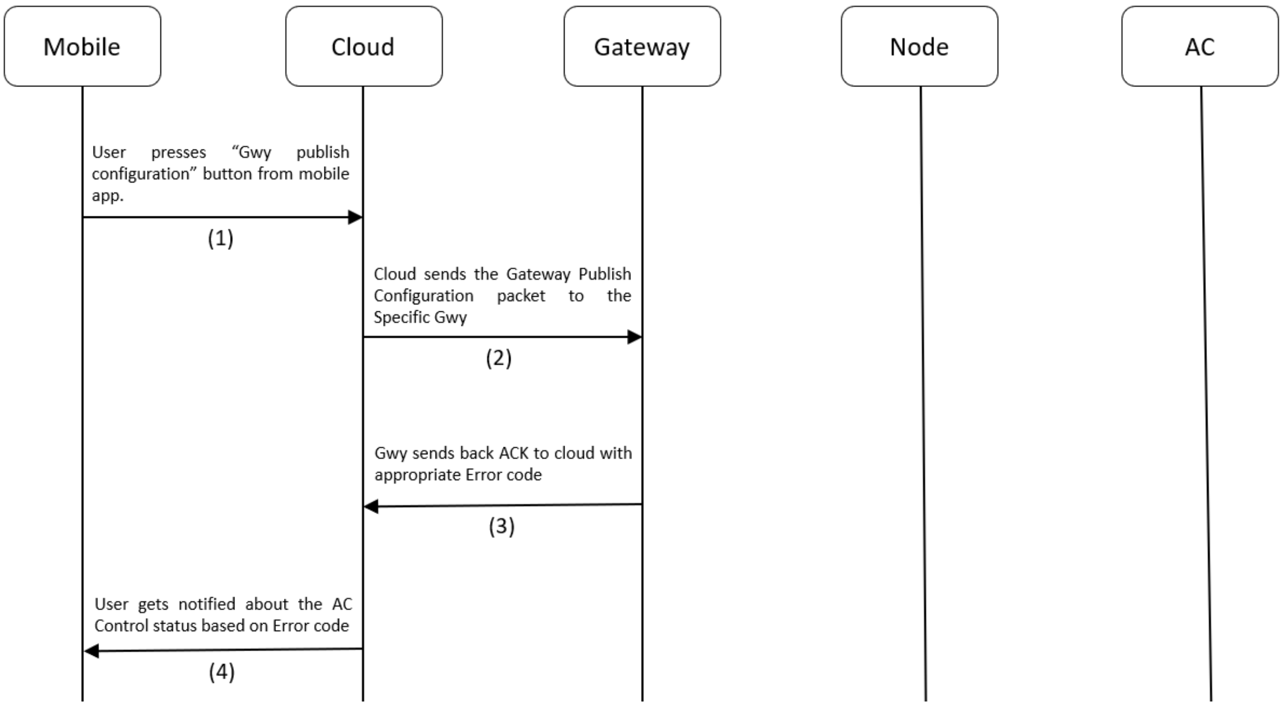


Figure 14 Gateway Publish Configuration Process flow

Gateway Reset MQTT Packet

Description: This packet is used to reset the MQTT parameters saved in the Gateway. This puts the Gateway into MQTT configuration mode.

Note: This will not Unregister or Unconfigure Gateway from the BLE Mesh network. But until Gateway is provided with new MQTT parameters, it will not be able to receive any more commands from Cloud. While in this state, Gateway – Node communciation will still be unaffected. But this state should not be maintained for longer periods, as Gateway will keep accumulating data received from Nodes and unable to send them out anywhere. Soon, buffer will get filled and data will start to get overwritten leading to data loss.

Gateway Reset MQTT JSON (Cloud to Gateway):

				Range
{				
"JsonPacketId"	:	8	// (Number)	
"MsgSeqNo"	:	12	// (Number)	[0 – 65535]
"GwySerNo"	:	"GWY00001"	// (String)	[00001 - 99999]
}				

Process flow :

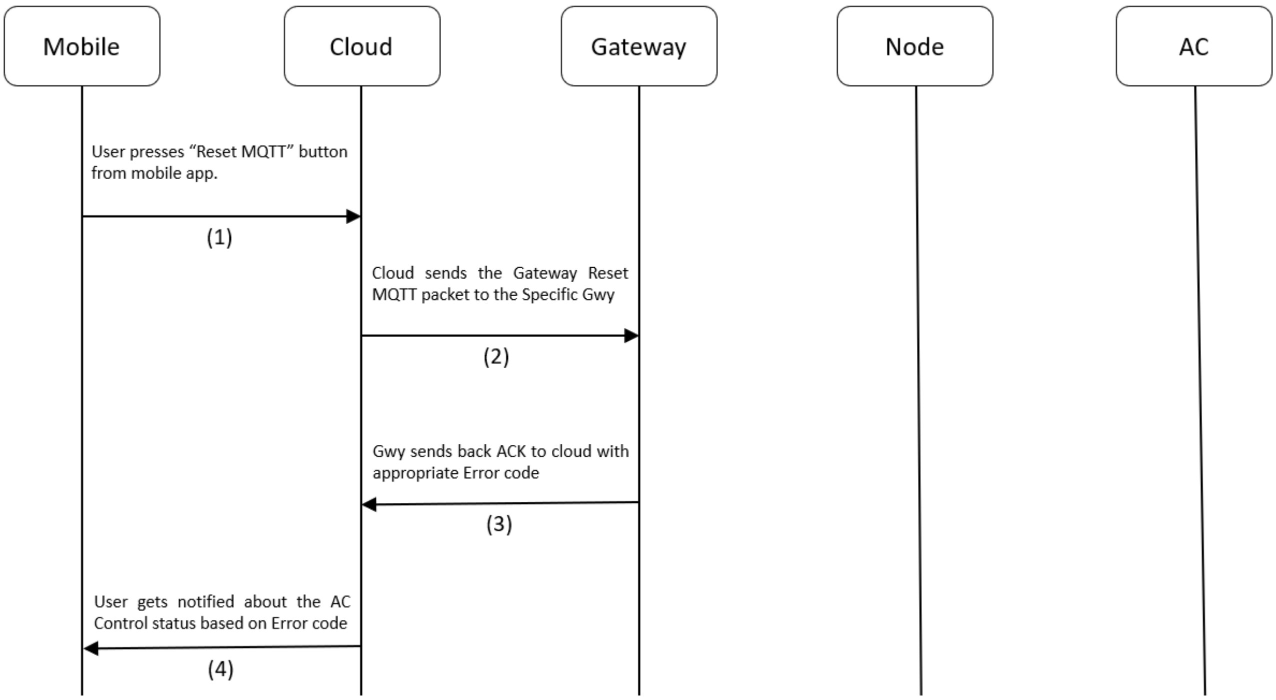


Figure 15 Gateway Reset MQTT Process flow

Gateway Reset MQTT Ack

Description: This ACK is used by cloud to know whether the Reset MQTT operation was successful or not.

Gateway Reset MQTT JSON ACK (Gateway to Cloud):


```
{
    “JsonPacketId”   :   8              // (Number)
    “GwySerNo”       :   “GWY00001”     // (String)
    “ErrorCode”      :   0              // (Number)
}
```


Process flow :

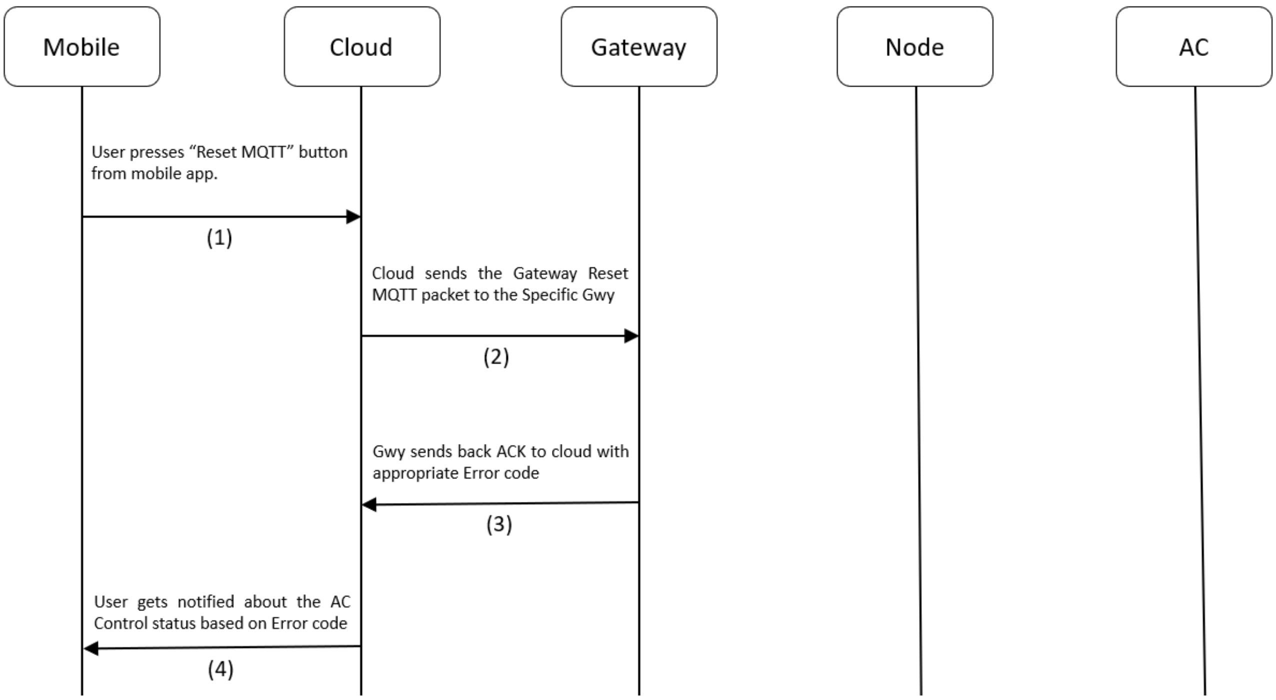


Figure 16 Gateway Reset MQTT Process flow

Gateway Teaching Mode Start Packet

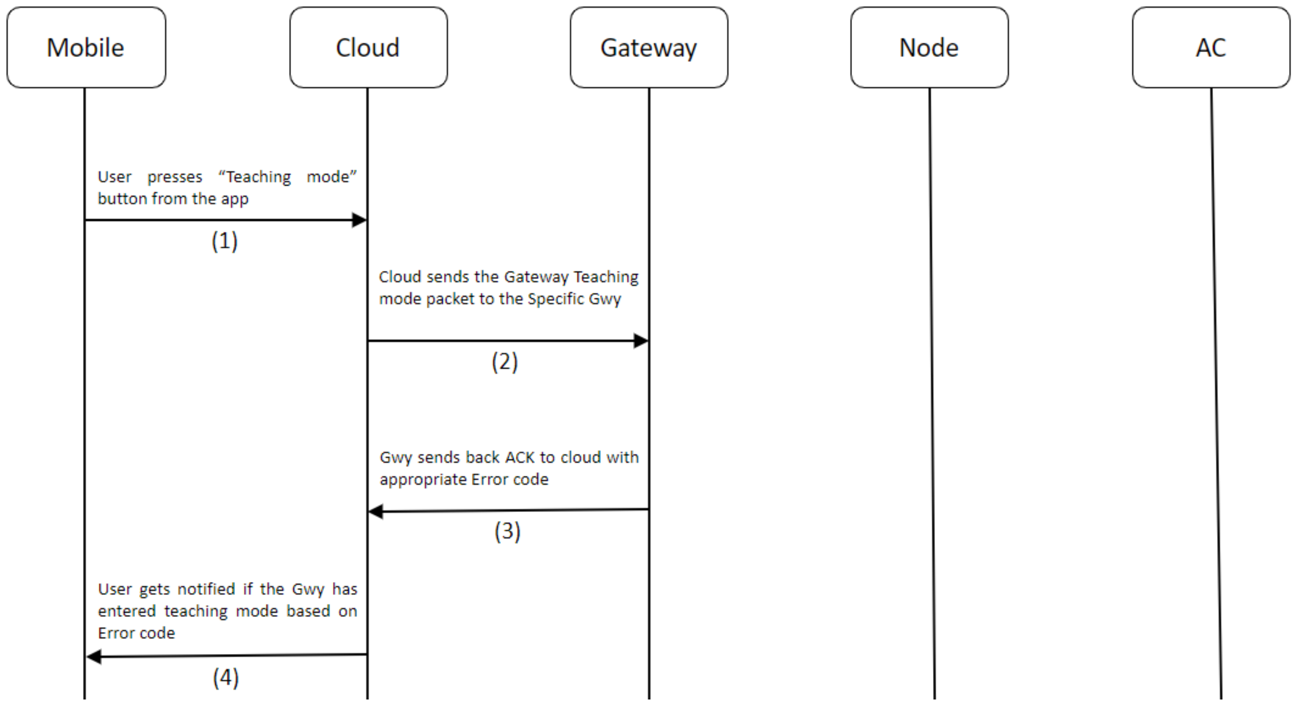
Description: This packet is used to make a Gateway enter Teaching mode.

Led Indication: Fast Blinking BLUE (200ms ON | 200ms OFF).

Gateway Teaching Mode Start JSON (Cloud to Gateway):

				Range
{				
"JsonPacketId"	:	9	// (Number)	
"MsgSeqNo"	:	12	// (Number)	[0 – 65535]
"GwySerNo"	:	"GWY00001"	// (String)	[00001 - 99999]
"StartingTemp"	:	20	// (Number)	[16 - 32]
"EndingTemp"	:	27	// (Number)	[16 - 32]
}				

Process flow:



Gateway Teaching Mode Start flow

Gateway Teaching Mode Start Ack

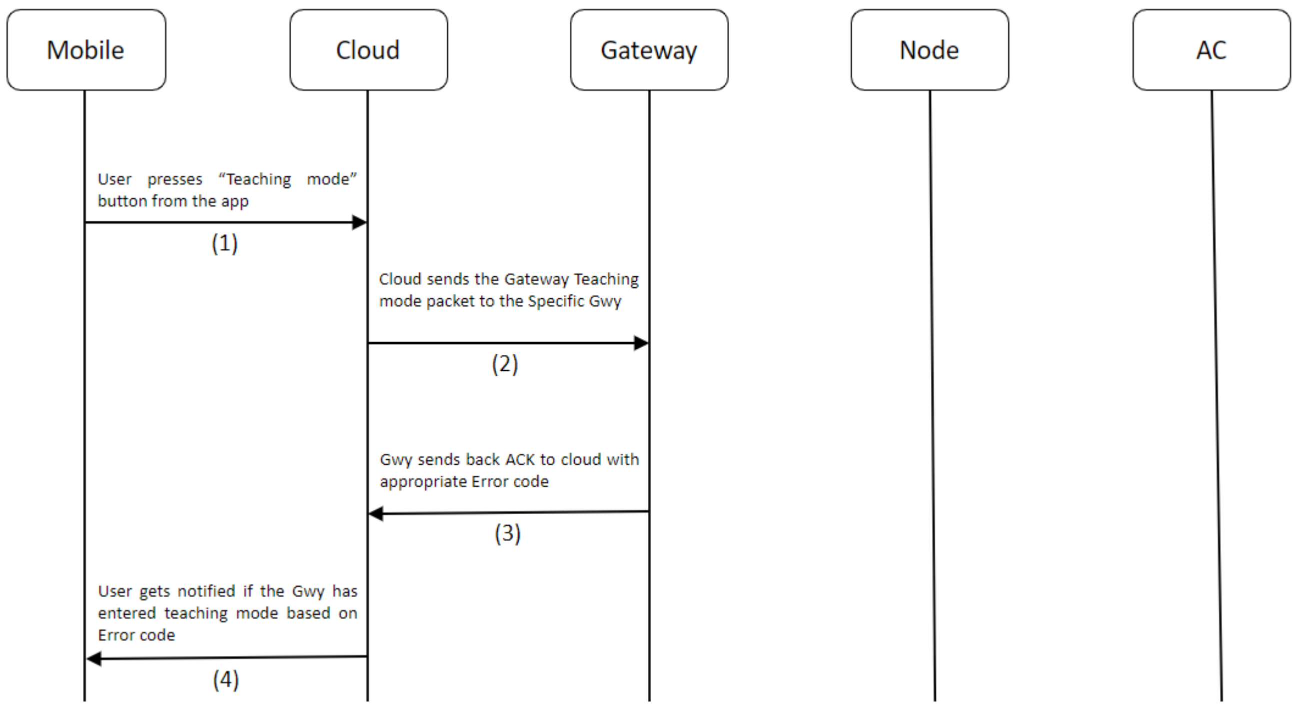
Description: This ACK is used by the cloud to know if the Gwy has successfully entered the teaching mode or not after receiving Gateway Teaching Mode Start Packet.

Led Indication: Fast Blinking BLUE (200ms ON | 200ms OFF).

Gateway Teaching Mode Start ACK JSON (Gateway to Cloud):


```
{
    “JsonPacketId”    :  9             // (Number)
    “MsgSeqNo”        :  12            // (Number)
    “GwySerNo”        :  “GWY00001”    // (String)
    “StartingTemp”    :  20            // (Number)
    “EndingTemp”      :  27            // (Number)
    “ErrorCode”       :  0             // (Number)
}
```


Process flow:



Gateway Teaching Mode Start flow

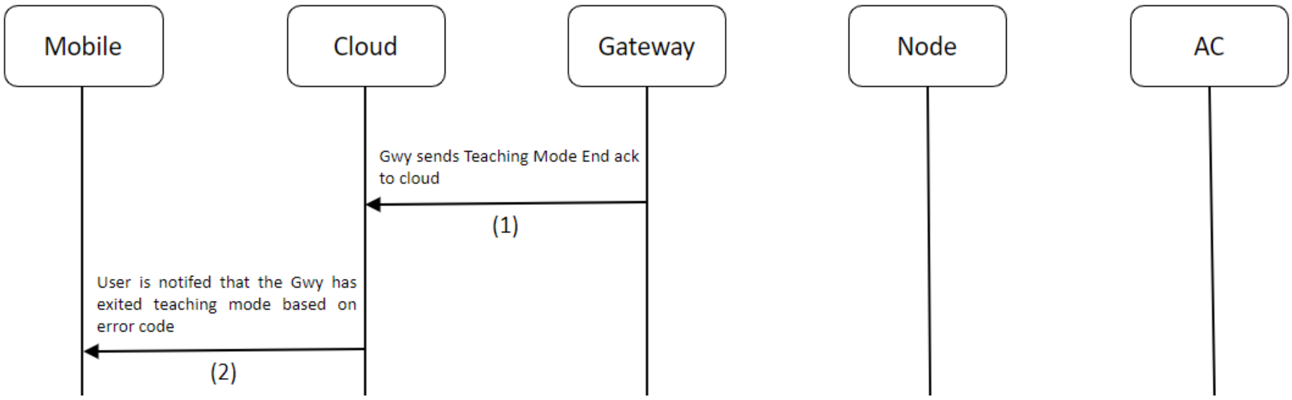
Gateway Teaching Mode End Ack

Description: This ACK is used by the cloud to know if the Gwy has exited teaching mode.

Gateway Teaching Mode End ACK JSON (Gateway to Cloud):


```
{
    "JsonPacketId"    :   10              // (Number)
    "MsgSeqNo"        :   12              // (Number)
    "GwySerNo"        :   "GWY00001"      // (String)
    "StartingTemp"    :   20              // (Number)
    "EndingTemp"      :   27              // (Number)
    "ErrorCode"       :   0               // (Number)
}
```


Process flow:



Gateway Teaching Mode End flow

Node Provision Packet

Description: This packet is used to add a Node device into the BLE Mesh network

Node Provisioning JSON (Cloud to Gateway to Node):

```
{                                                                          Range
    “JsonPacketId”   :   100                       // (Number)
    “MsgSeqNo”       :   12                        // (Number)            [0 – 65535]
    “GwySerNo”       :   “GWY00001”                // (String)           [00001 - 99999]
    “NodeSerNo:      :   “N00001”                  // (String)           [00001 - 99999]
    “MacId”          :   “D3:F5:FB:CA:14:25”       // (String)           [17 chars long]
}
```

Process flow :

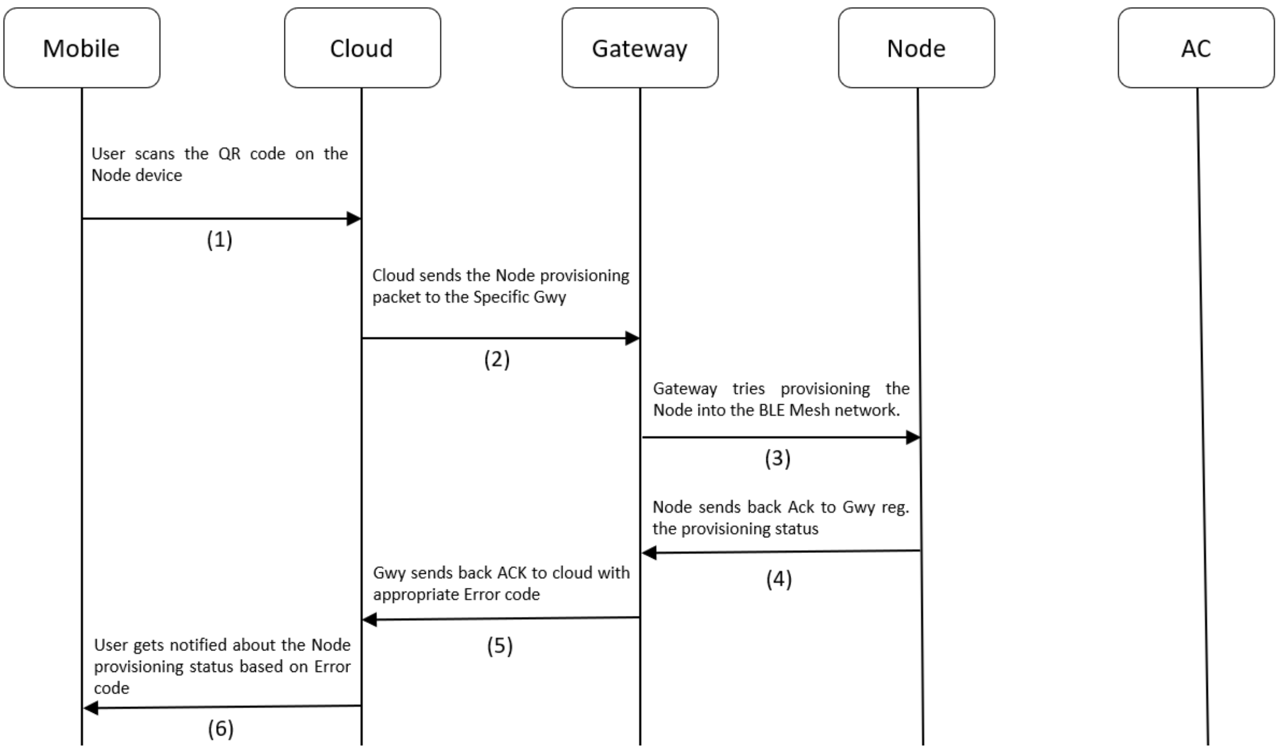


Figure 17 Node Provisioning Process flow

Node Provision Ack

Description: This ack is used by cloud to know whether the Node provisioning was successful or not.

Note: In this ACK, the **ElementAddr** is a very important field, whose value should be noted for further Node operations. Ony using this ElementAddr, the Gateway will be able to communicate with the Node. This ElementAddr field will be included in all Node related MQTT packets.

Node Provisioning JSON ACK (Node to Gateway to Cloud):

```
{                                                                    Range
    “JsonPacketId”   :  100                  // (Number)
    “MsgSeqNo”       :  12                   // (Number)
    “GwySerNo”       :  “GWY00001”           // (String)
    “NodeSerNo:      :  “N00001”             // (String)
    “MacId”          :  “D3:F5:FB:CA:14:25”  // (String)
    “ElementAddr”    :  23                   // (Number)          [2 - 65535]
    “ErrorCode”      :  0                    // (Number)
}
```

Process flow :

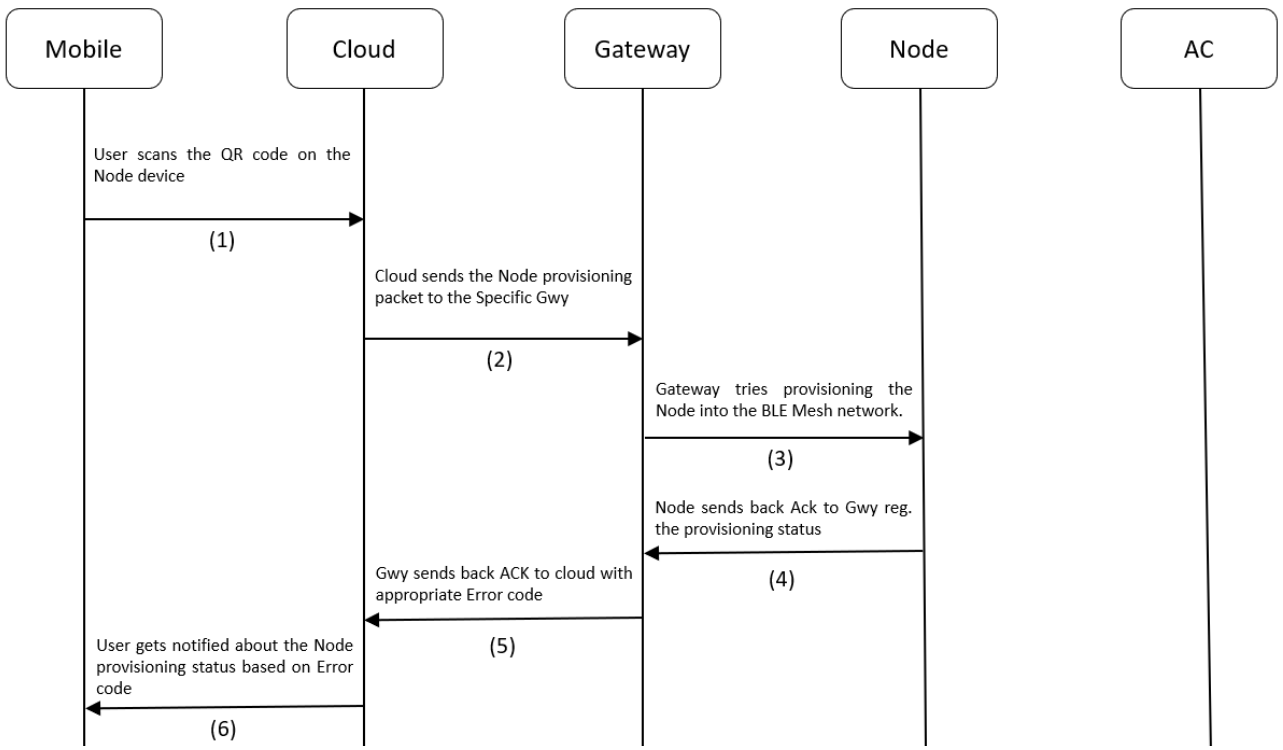


Figure 18 Node Provisioning Process flow

Node AC Remote Configuration Ack

Description: This is an ACK sent by Node to Gwy to Cloud when the Node is successfully configured with an AC remote. When Node is in AC Remote Configuration mode and AC remote is pressed in-front of the IR receiver in the Node, it will get configured to act as that AC remote from there on.

LED Indication: Slow Blinking BLUE (500ms ON | 500ms OFF)

Node Configuration JSON ACK (Node to Gateway to Cloud):

				Range
{				
"JsonPacketId"	:	101	// (Number)	
"MsgSeqNo"	:	12	// (Number)	
"GwySerNo"	:	"GWY00001"	// (String)	
"NodeSerNo:	:	"N00001"	// (String)	
"ElementAddr"	:	23	// (Number)	[2 - 65535]
"ErrorCode"	:	0	// (Number)	
}				

Process flow :

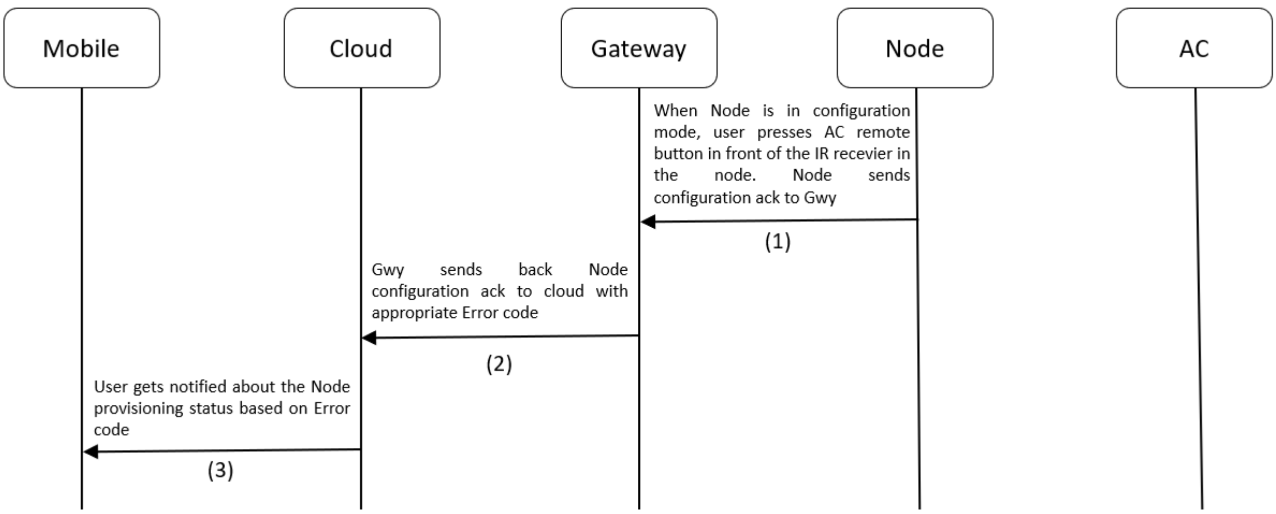


Figure 19 Node Configuration Process flow

Node Unprovision Packet

Description: This packet is used to remove a Node device from the BLE Mesh network.

Node Unprovisioning JSON (Cloud to Gateway to Node):

```
{                                                                                Range
    "JsonPacketId"   :  102              // (Number)
    "MsgSeqNo"       :  12               // (Number)
    "GwySerNo"       :  "GWY00001"       // (String)
    "NodeSerNo:      :  "N00001"         // (String)
    "ElementAddr"    :  23               // (Number)                       [2 - 65535]
}
```

Process flow :

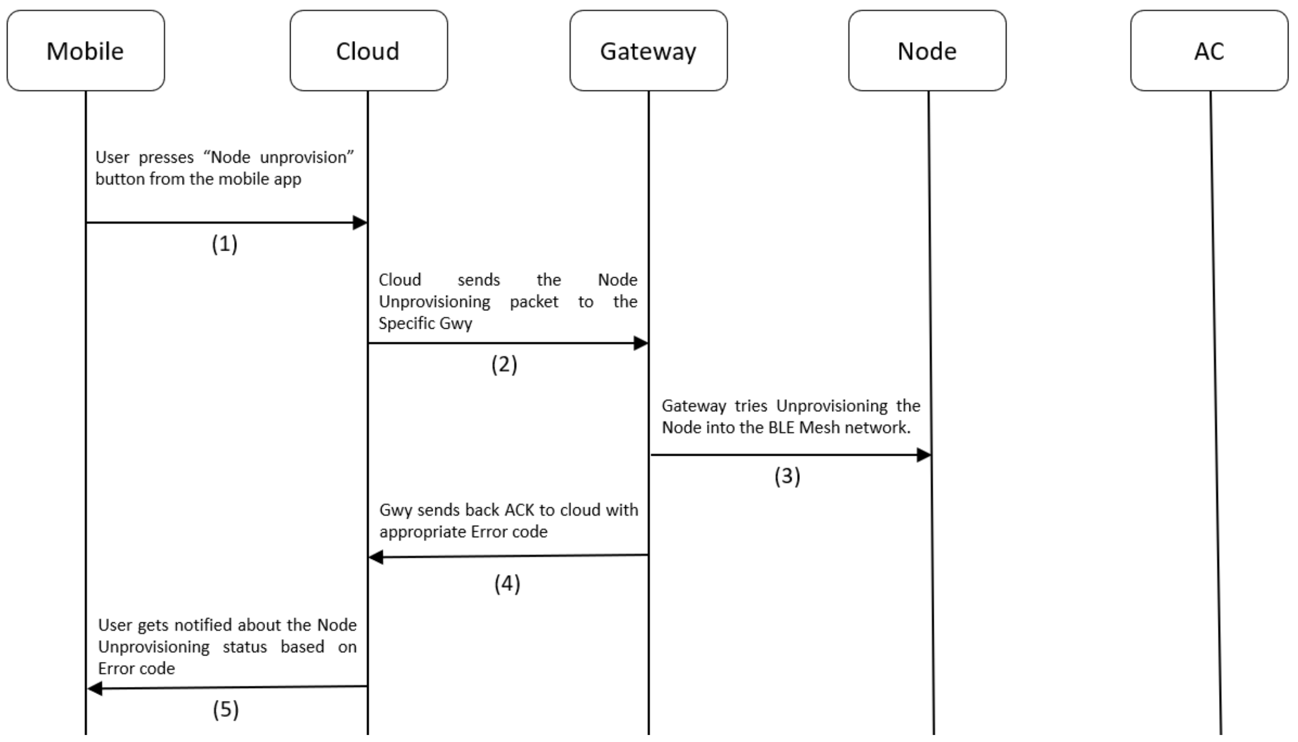


Figure 20 Node Unprovsioning Process flow

Node Unprovision Ack

Description: This packet is used to add a Node device into the BLE Mesh network

Node Unprovisioning JSON ACK (Node to Gateway to Cloud):

```
{                                                                       Range
    "JsonPacketId"  :   102             // (Number)
    "MsgSeqNo"      :   12              // (Number)
    "GwySerNo"      :   "GWY00001"      // (String)
    "NodeSerNo:     :   "N00001"        // (String)
    "ElementAddr"   :   23              // (Number)              [2 - 65535]
    "ErrorCode"     :   0               // (Number)
}
```

Process flow :

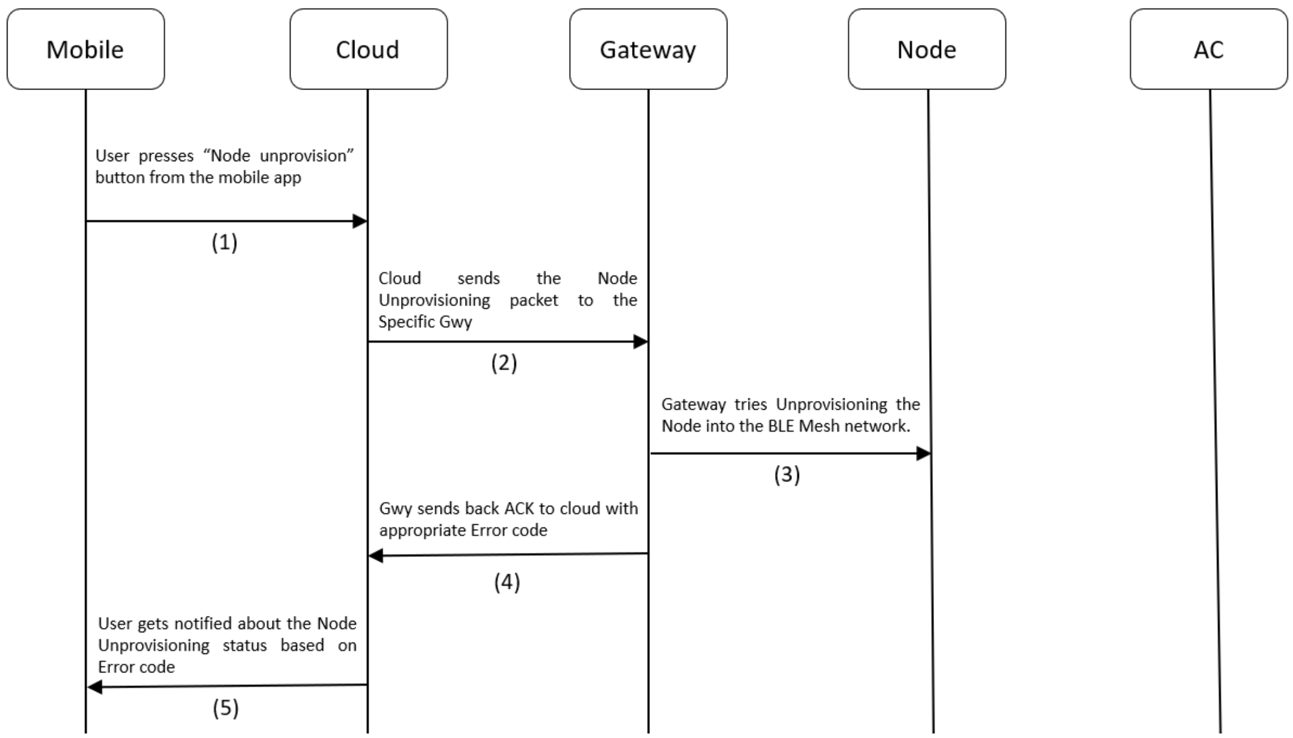


Figure 21 Node Unprovisioning Process flow

Node AC Control Packet

Description: This packet is used to control an AC using Node.

Table 0.6 Node AC Control JSON Parameters

Sl.No	Parameter	Description	Possible Err Codes
1	Power	1 = ON, 0 = OFF	INVALID_POWER
2	Mode	Following modes are supported : [Cool, Dry, Hot, Fan, Auto].	INVALID_MODE
3	Temperature	Sets the AC temperature. Must be within 18 and 32.	EXCEEDING_TEMP_LOWER_LIMIT, EXCEEDING_TEMP_UPPER_LIMIT
4	SwingH	1 = ON, 0 = OFF	INVALID_SWING_H
5	SwingV	1 = ON, 0 = OFF	INVALID_SWING_V
6	OnTimer	Sets the number of hours past midnight after which AC needs to be turned on automatically	INVALID_ONTIMER
7	OffTimer	Sets the number of hours past midnight after which AC needs to be turned off automatically	INVALID_OFFTIMER
8	Locking	1 = ON, 0 = OFF. When ON, if AC was controlled manually, then Locking ACK will be sent to Cloud. Also, when ON, the AC will be maintained between the TempLockUpLimit and TempLockLowLimit. If, manual control, AC temp was set exceeding the above limits, then AC's temperature will be brought back under accepted limits.	INVALID_LOCKING
9	TempLockUpLimit	Specifies upper temperature locking limit. This parameter will have no effect application wise, when Locking is not enabled. Must have values within the absolute Temperature limits : 18 - 32. Default value : 27	INVALID_TEMP_UPPER_LIMIT, ILLOGICAL_LOCKING_TEMP_LIMIT
10	TempLockLowLimit	Specifies Lower temperature locking limit. This parameter will have no effect application wise, when Locking is not enabled. Must have values	INVALID_TEMP_LOWER_LIMIT, ILLOGICAL_LOCKING_TEMP_LIMIT

		within the absolute Temperature limits : 18 - 32. Default value : 20.	

Node AC Control JSON (Cloud to Gateway to Node):

```
{                                                                    Range
    "JsonPacketId"       :  103           // (Number)
    "MsgSeqNo"           :  12            // (Number)    [0 - 65535]
    "GwySerNo"           :  "GWY00001"    // (String)    [00001 - 99999]
    "NodeSerNo"          :  "N00001"      // (String)    [00001 - 99999]
    "ElementAddr"        :  23            // (Number)    [2 - 65535]
    "Power"              :  1             // (Number)    [0 = OFF, 1 = ON]
    "Mode"               :  "Cool"        // (String)    ["Cool", "Dry", "Auto", "Fan", "Hot"]
    "Fan"                :  1             // (Number)    [1 - 5]
    "Temperature"        :  25            // (Number)    [18 - 32]
    "SwingH"             :  1             // (Number)    [0 = OFF, 1 = ON]
    "SwingV"             :  1             // (Number)    [0 = OFF, 1 = ON]
    "OnTimer"            :  0             // (Number)    [0 - 12]
    "OffTimer"           :  0             // (Number)    [0 - 12]
    "Locking"            :  1             // (Number)    [0 = OFF, 1 = ON]
    "TempLockUpLimit"    :  26            // (Number)    [18 - 32]
    "TempLockLowLimit"   :  20            // (Number)    [18 - 32]
}
```

Process flow :

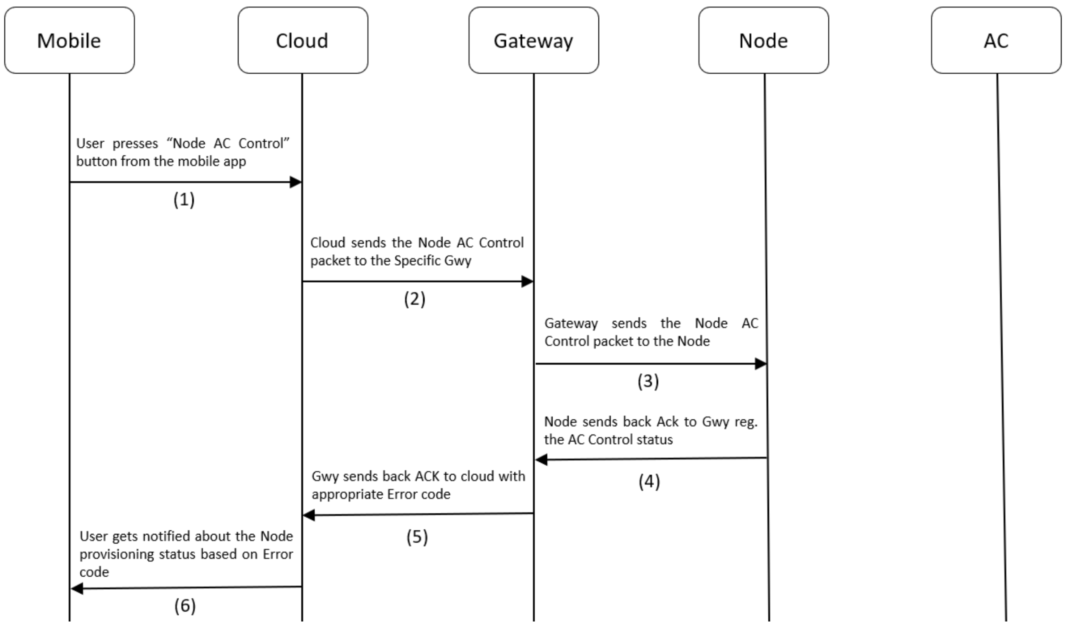


Figure 22 Node AC Control Process flow

Node AC Control Ack

Description: This is used by cloud to know if the AC Control was successful or not.

Node AC Control JSON ACK (Node to Gateway to Cloud):


```
{
    "JsonPacketId"      :  103           // (Number)
    "MsgSeqNo"          :  12            // (Number)
    "GwySerNo"          :  "GWY00001"    // (String)
    "NodeSerNo"         :  "N00001"      // (String)
    "ElementAddr"       :  23            // (Number)
    "Power"             :  1             // (Number)
    "Mode"              :  "Cool"        // (String)
    "Fan"               :  1             // (Number)
    "Temperature"       :  25            // (Number)
    "SwingH"            :  1             // (Number)
    "SwingV"            :  1             // (Number)
    "OnTimer"           :  0             // (Number)
    "OffTimer"          :  0             // (Number)
    "Locking"           :  1             // (Number)
    "TempLockUpLimit"   :  26            // (Number)
    "TempLockLowLimit"  :  20            // (Number)
    "ErrorCode"         :  0             // (Number)
}
```


Process flow :

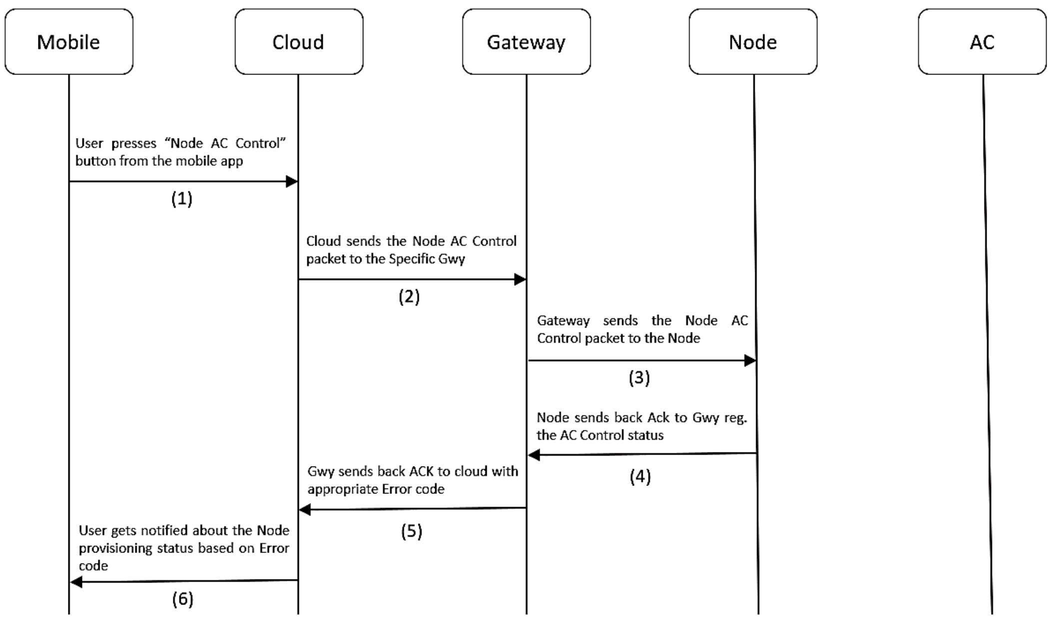


Figure 0.24 Node AC Control Process flow

Node AC manual control Ack

Description: This ack is used by cloud to know whether the AC was controlled manually using the AC remote. If someone controls the AC manually using the remote, then the IR receiver on-board, will capture this IR signal and decode it. If it's a signal from the same remote with which the Node was configured with during the AC Remote configuration process, then it will store the manual control information for sending it to the cloud. Also it performs logical operation based on the temperature that was set using the remote. If it exceeds the Locking temperature limits set from the AC Control packet, then it will bring the AC's temperature back to permissible limits.
Note: This ACK will be sent from Node to Gateway to Cloud only if the Locking feature had been enabled. By default, the TempLockUpLimit is set to 27 and TempLockLowLimit is set to 20

Node AC Locking JSON ACK (Node to Gateway to Cloud):

```
{                                                                        Range
    "JsonPacketId"   :  104              // (Number)
    "MsgSeqNo"       :  12               // (Number)
    "GwySerNo"       :  "GWY00001"       // (String)
    "NodeSerNo:      :  "N00001"         // (String)
    "ElementAddr"    :  23               // (Number)              [2 - 65535]
    "Power"          :  1                // (Number)
    "Mode"           :  "Cool"           // (String)
    "FanSpeed"       :  1                // (Number)
    "Temperature"    :  25               // (Number)
    "SwingH"         :  1                // (Number)
    "SwingV"         :  1                // (Number)
    "OnTimer"        :  0                // (Number)
    "OffTimer"       :  0                // (Number)
}
```

Process flow :

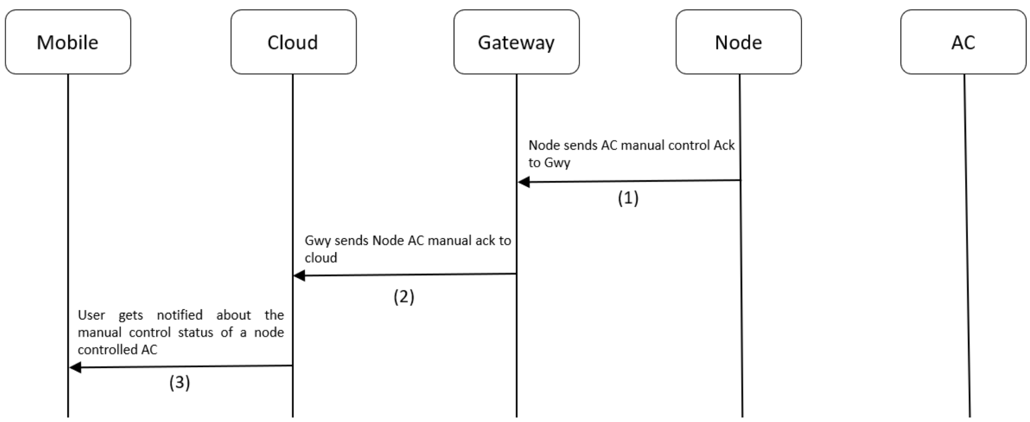


Figure 23 Node AC Locking Process flow

Node AC Remote Reconfiguration Packet

Description: This packet is used to reset the MQTT parameters saved in the Gateway. This puts the Node into AC Remote Configuration mode.

LED Indication: Slow Blinking BLUE (500ms ON | 500ms OFF)

Node AC Remote Reconfiguration JSON (Cloud to Gateway to Node):

{				**Range**
"JsonPacketId"	:	105	// (Number)	
"MsgSeqNo"	:	12	// (Number)	[0 – 65535]
"GwySerNo"	:	"GWY00001"	// (String)	[00000 - 99999]
"NodeSerNo:	:	"N00001"	// (String)	
"ElementAddr"	:	23	// (Number)	
}				

Process flow :

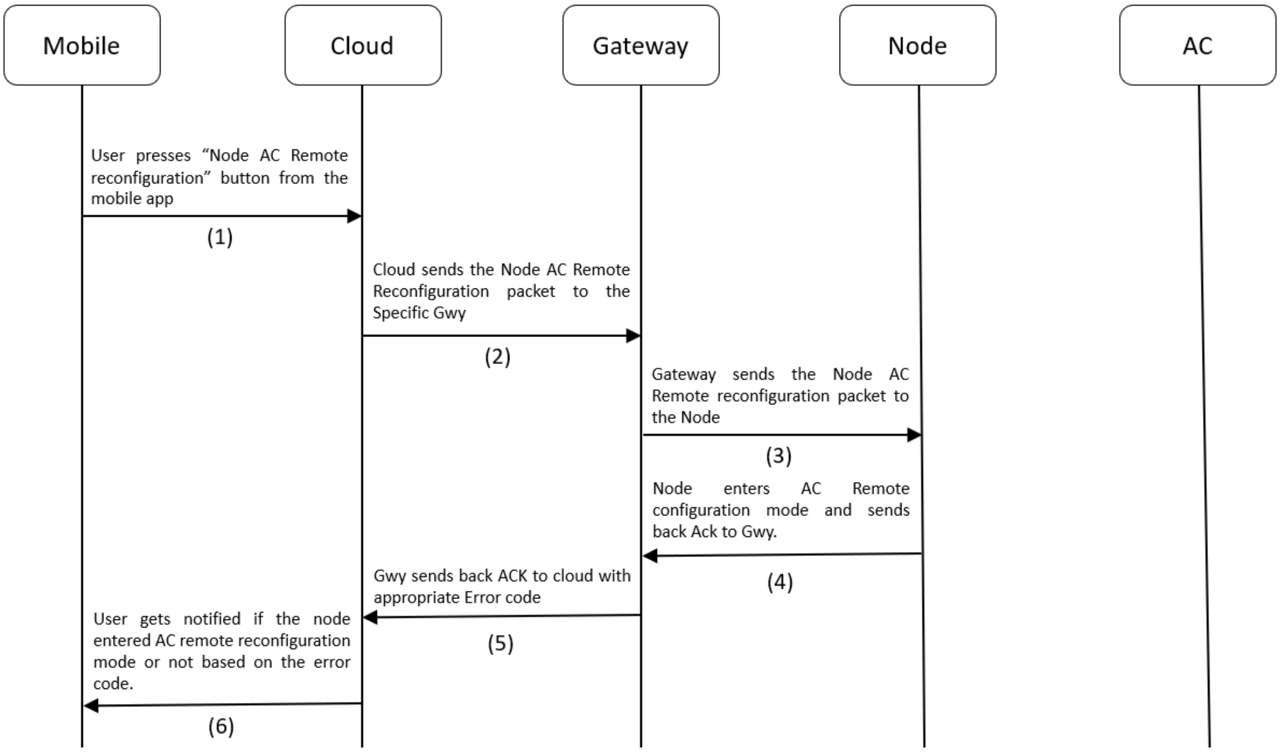


Figure 24 Node Reconfiguration Process flow

Node AC Remote Reconfiguration Ack

Description: This ack is used by cloud to know whether the node succesfully entered AC Remote configuration mode or not, after receving the Node AC Remote Reconfiguration Packet.

LED Indication: Slow Blinking BLUE (500ms ON | 500ms OFF)

Node Reconfiguration JSON ACK (Node to Gateway to Cloud):

```
{                                                                  Range
    “JsonPacketId”   :   105              // (Number)
    “MsgSeqNo”       :   12               // (Number)
    “GwySerNo”       :   “GWY00001”       // (String)
    “NodeSerNo:      :   “N00001”         // (String)
    “ElementAddr”    :   23               // (Number)            [2 - 65535]
    “ErrorCode”      :   0                // (Number)
}
```

Process flow :

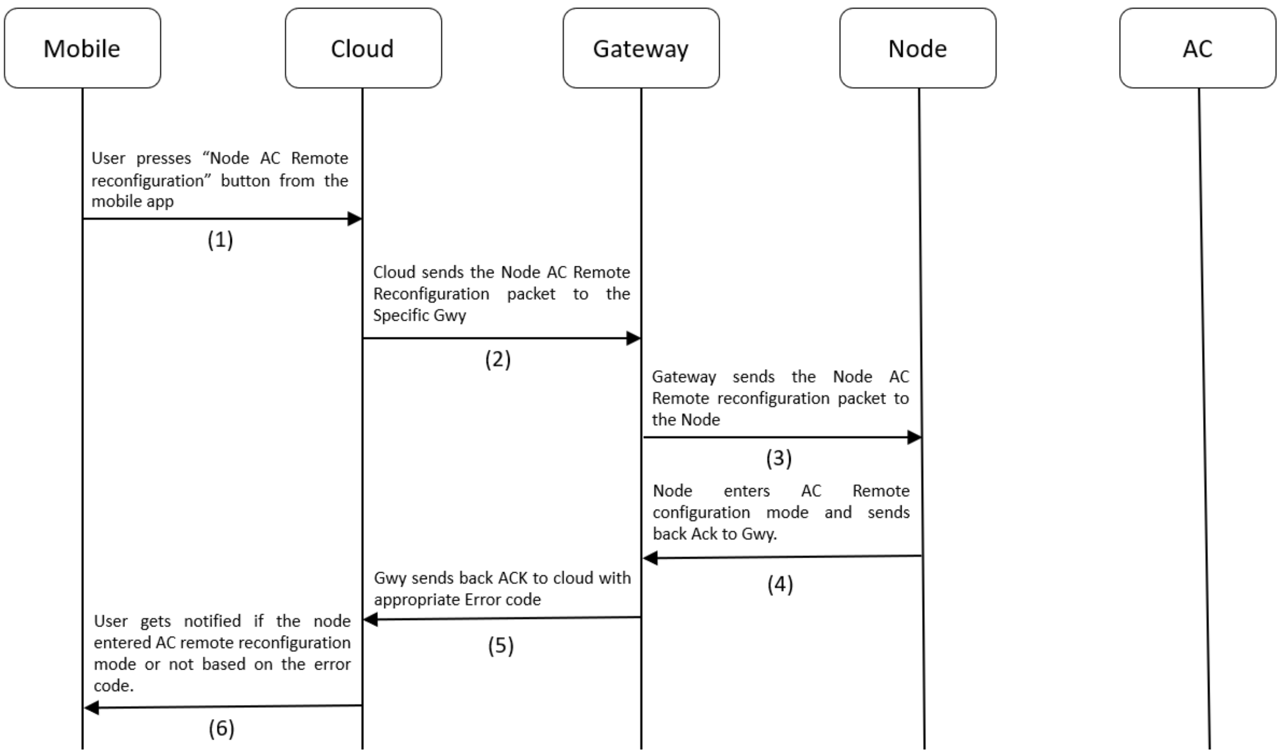


Figure 25 Node Reconfiguration Process flow

Node Temperature Data Ack

Description: This ACK is used by cloud to know the ambient temperature of the environment in which the Node is operating. This ACK is sent periodically to the Gwy to Cloud. This frequency can be configured in the Node Temperature Publish Configuration Packet

Note: This ACK is also used for Heartbeat purpose.

Node Temperature Data JSON ACK (Node to Gateway to Cloud):

```
{                                                                        Range
     "JsonPacketId"   :   106              // (Number)
     "MsgSeqNo"       :   12               // (Number)
     "GwySerNo"       :   "GWY00001"       // (String)
     "NodeSerNo:      :   "N00001"         // (String)
     "ElementAddr"    :   23               // (Number)                 [2 - 65535]
     "Temperature"    :   26               // (Number)
}
```

Process flow :

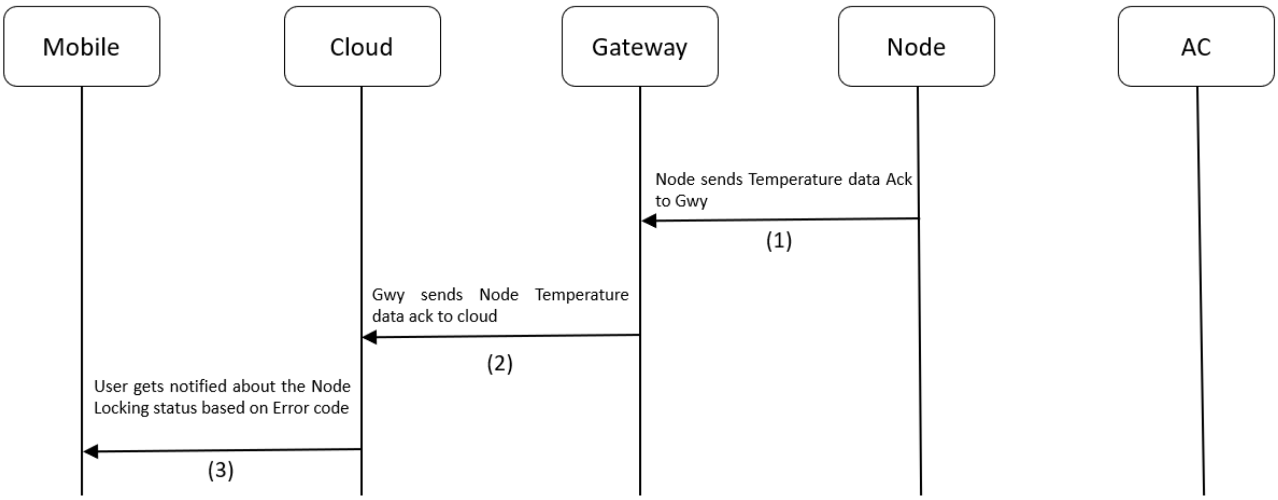


Figure 26 Node Temperature Data Process flow

Node Temperature Publish Configuration Packet

Description: This packet is used to tell the Node, how frequently It needs to send Temperature data ack to Gwy, so that Gwy can send it to Cloud.

Node Temperature Publish Configuration JSON (Cloud to Gateway):


```
{                                                                                 Range
    "JsonPacketId"      :  108              // (Number)
    "MsgSeqNo"          :  12               // (Number)                  [0 – 65535]
    "GwySerNo"          :  "GWY00001"       // (String)                  [00001 - 99999]
    "NodeSerNo"         :  "N00001"         // (String)                  [00001 - 99999]
    "ElementAddr"       :  23               // (Number)                  [2 - 65535]
    "PublishPeriodSec"  :  15               // (Number) (in sec)         [5 - 255]
}
```


Process flow :

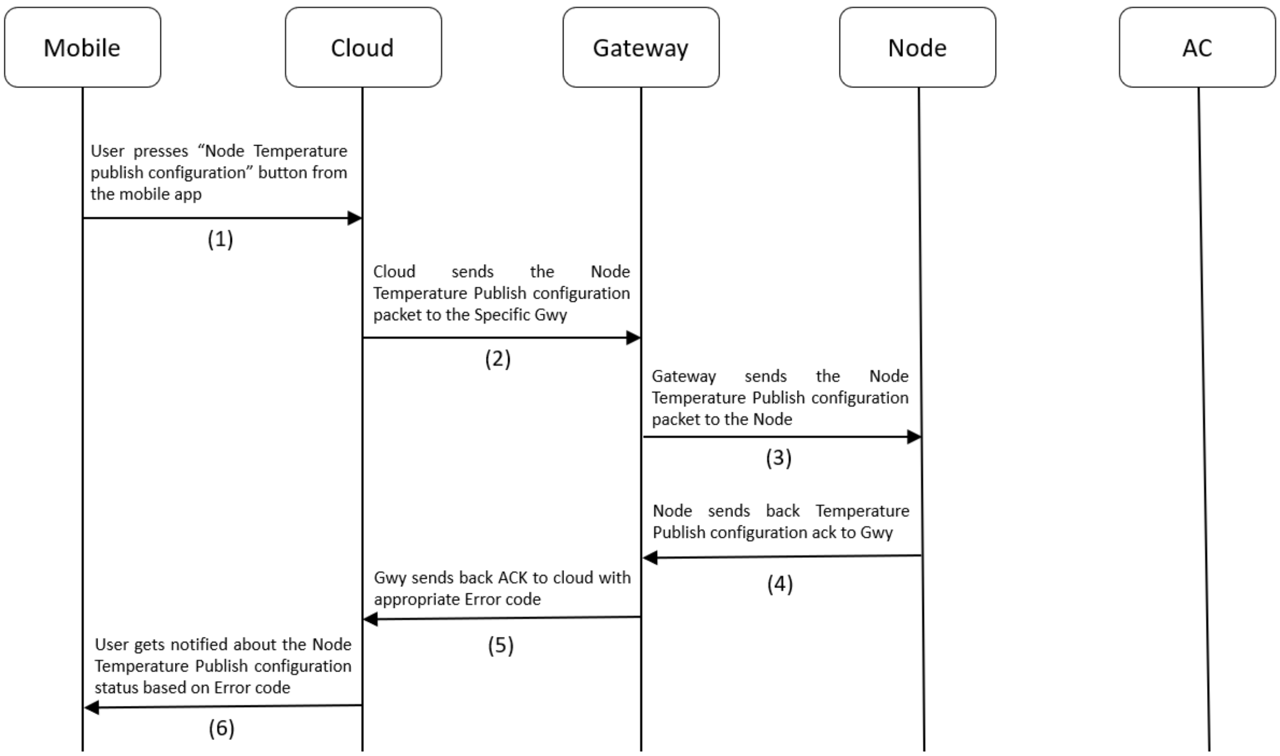


Figure 27 Node Publish Configuration process flow

Node Temperature Publish Configuration Ack

Description: This ack is used by cloud to know if the Node publish configuration was successful or not.

Node Publish Configuration JSON ACK (Node to Gateway to Cloud):

{				**Range**
"JsonPacketId"	:	108	// (Number)	
"MsgSeqNo"	:	12	// (Number)	
"GwySerNo"	:	"GWY00001"	// (String)	
"NodeSerNo:	:	"N00001"	// (String)	
"ElementAddr"	:	23	// (Number)	[2 - 65535]
"PublishPeriodSec"	:	15	// (Number)	
"ErrorCode"	:	0	// (Number)	
}				

Process flow :

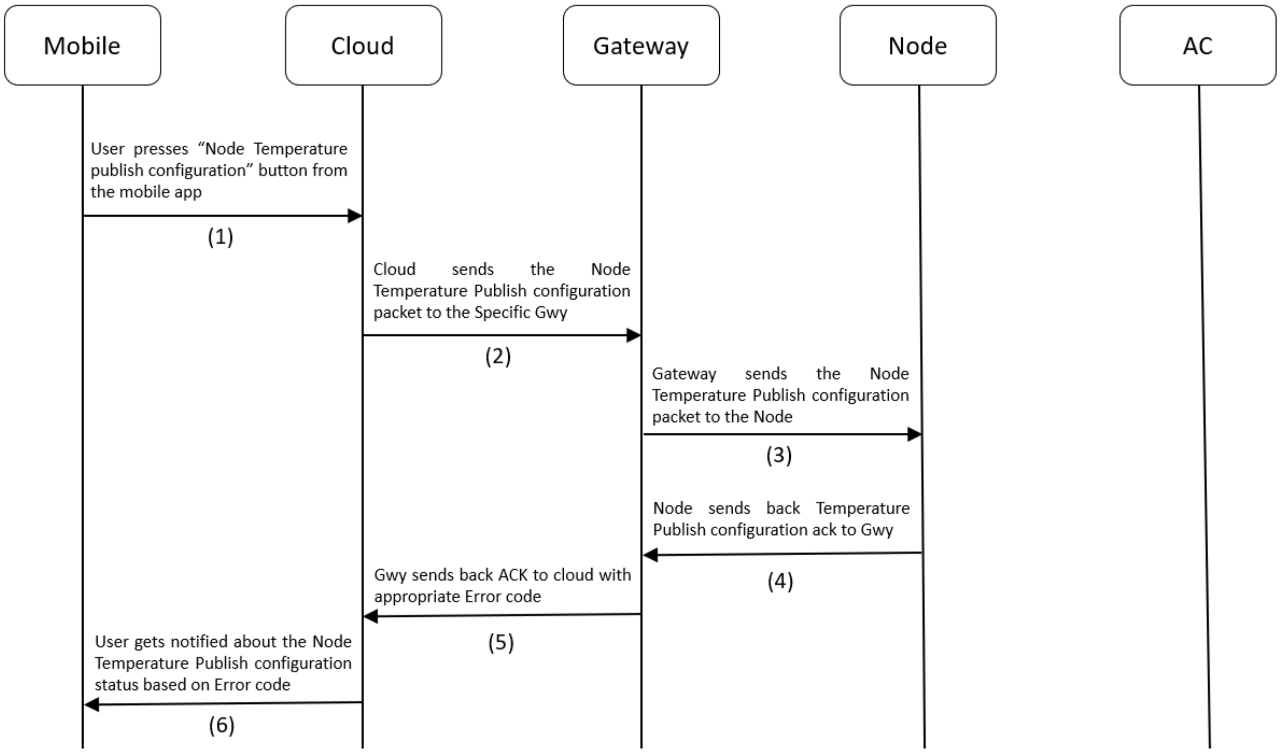


Figure 28 Node Publish Configuration process flow

Node Teaching Mode Start Packet

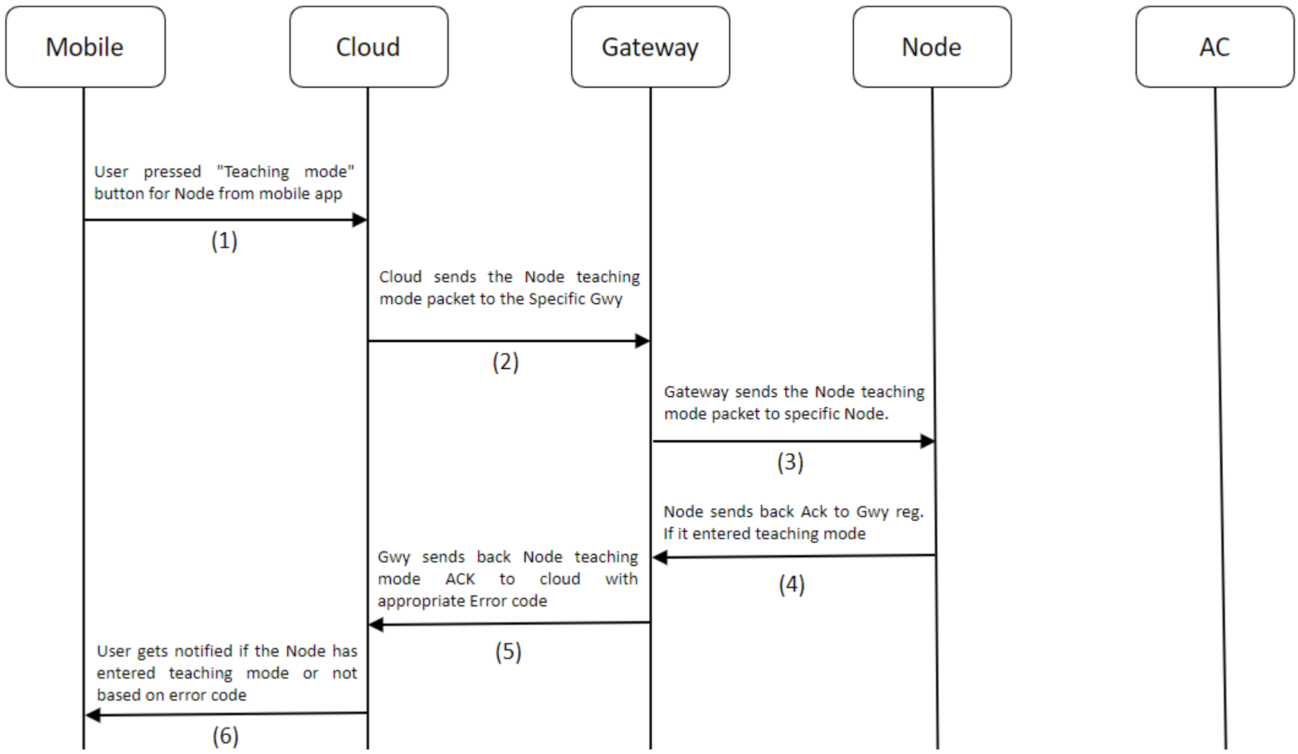
Description: This packet is used to make a Node enter Teaching mode.

Led Indication: Fast Blinking BLUE (200ms ON | 200ms OFF).

Node Teaching Mode Start JSON (Cloud to Gateway to Node):

```
{                                                                    Range
    “JsonPacketId”   :  108          // (Number)
    “MsgSeqNo”       :  12           // (Number)              [0 – 65535]
    “GwySerNo”       :  “GWY00001”   // (String)             [00001 - 99999]
    “NodeSerNo”      :  “N00001”     // (String)             [00001 - 99999]
    “ElementAddr”    :  23           // (Number)              [2 - 65535]
    “StartingTemp”   :  20           // (Number)               [16 - 32]
    “EndingTemp”     :  27           // (Number)               [16 - 32]
}
```

Process flow:



Node Teaching Mode Start flow

Node Teaching Mode Start Ack

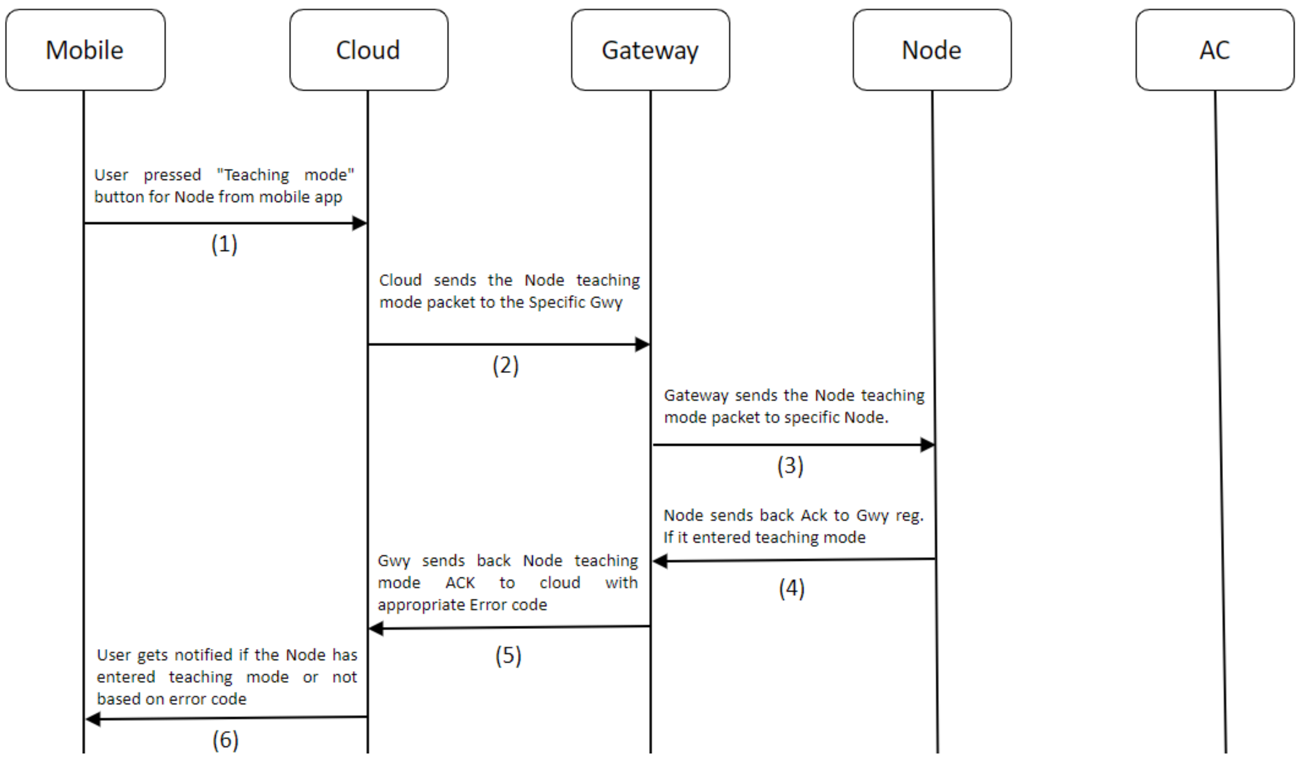
Description: This ACK is used by the cloud to know if the Node has successfully entered the teaching mode or not after receiving Node Teaching Mode Start Packet.

Led Indication: Fast Blinking BLUE (200ms ON | 200ms OFF).

Node Teaching Mode start ACK JSON (Node to Gateway to Cloud):


```
{
    "JsonPacketId"    :   108           // (Number)
    "MsgSeqNo"        :   12            // (Number)
    "GwySerNo"        :   "GWY00001"    // (String)
    "NodeSerNo"       :   "N00001"      // (String)
    "ElementAddr"     :   23            // (Number)
    "StartingTemp"    :   20            // (Number)
    "EndingTemp"      :   27            // (Number)
    "ErrorCode"       :   0             // (Number)
}
```


Process flow:



Node Teaching Mode Start flow

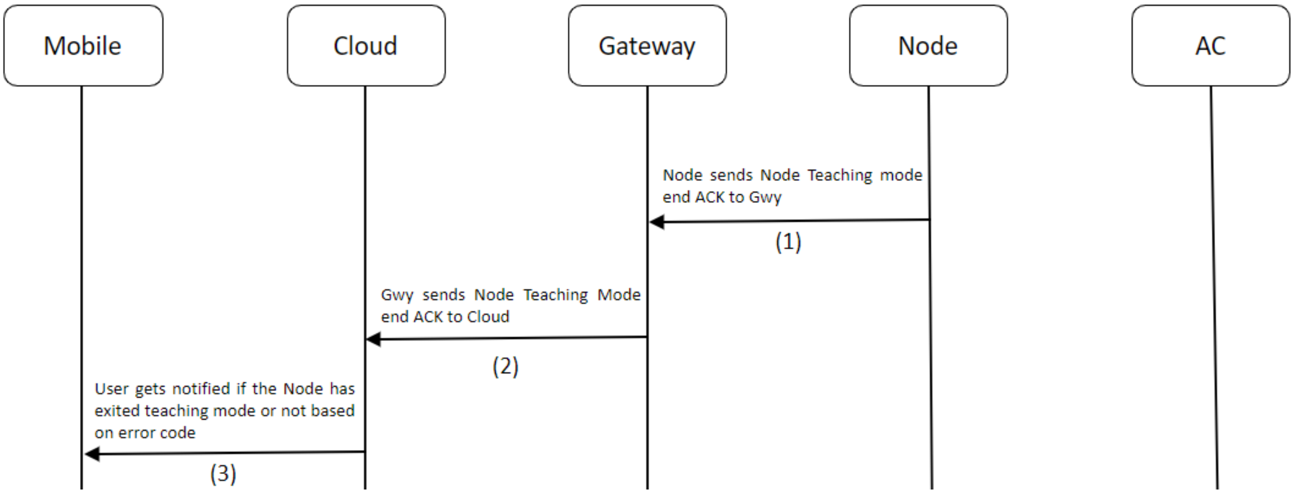
Node Teaching Mode End Ack

Description: This ACK is used by the cloud to know if the Node has exited teaching mode. Based on Error code user will know whether the teaching process was completed fully or not.

Node Teaching Mode End ACK JSON (Node to Gateway to Cloud):


```
{
    "JsonPacketId"   :  109           // (Number)
    "MsgSeqNo"       :  12            // (Number)
    "GwySerNo"       :  "GWY00001"    // (String)
    "StartingTemp"   :  20            // (Number)
    "EndingTemp"     :  27            // (Number)
    "ErrorCode"      :  0             // (Number)
}
```


Process flow:



Node Teaching Mode End flow

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